TESS ARCHIVAL FOLLOW-UP · 18-19 OCTOBER 2026 · CONDITIONAL RESULTS, NOTHING VALIDATED

Three TESS candidates: TIC 270573718, TIC 262843927, TIC 193718989

A skeptical, reproducible re-examination of three long-period signals from the S1 survey batch 1, using every accessible local product and public archive. No new observing epoch exists for any of them: no TESS sector after 98 contains these stars, so every result below re-reduces the discovery photons or tests them against external catalogues and ground photometry.

Summary · Inputs and provenance · Archive census · Methods · TIC 270573718 · TIC 262843927 · TIC 193718989 · Selection replay · Luminosity for colour · Synthesis · Test matrix · Failures and limits · Reproduction

Summary

TIC 262843927

P = 24.1089 d, 1.2-1.3 ppt, early M dwarf

Repeating astrophysical dip	HIGH
Named target is the host	MODERATE
Planet-sized, planet-mass body	MODERATE

Join the top tier, narrowly first. Likely unresolved companion: image it first.

TIC 193718989

P = 24.5329 d, 2.3 ppt, early M dwarf

Repeating astrophysical dip	HIGH
Named target is the host	MODERATE
Planet-sized, planet-mass body	LOW TO MODERATE

Host attribution improved. Join the top tier, close behind 262.

TIC 270573718

P = 41.3837 d, 1.7-2.1 ppt, early M dwarf

Repeating astrophysical dip	HIGH
Named target is the host	MODERATE
Planet-sized, planet-mass body	LOW

Retain in the second tier. Timing structure unresolved.

TIC 270573718. The earlier “corrected 0.29 versus aperture 8.19” contradiction reproduces exactly and is not a disagreement between data products. Only the

corrected series' early half spans 2018 and 2020. Its fitted period drift of -3.85 min per cycle, extrapolated over 45 cycles, moves the predicted late transits off the real ones. At the reference ephemeris the late events give S/N 7.25 (corrected) and 8.24 (aperture). Underneath it is a season-level timing pattern. All six 2020-21 transits arrive about 50-64 min early in every reduction, including two independent pixel extractions. Strictly periodic injections into the same data never produce that much coherence (tail $p \approx 0.04$ per reduction). A free-shape fit with physically bounded shapes reproduces the early 2020-21 arrival in both PDCSAP and pixel data (unbounded, it is weaker in PDCSAP and unstable in the pixel series). Heavy-tailed timing errors still keep the protocol's claim rule from being met, so a transit-timing variation is suggestive, not established. With the season offsets removed, the transit is flat-bottomed and consistent with the catalogue density. The grazing, $e \approx 0.87$ alternative was largely a product of folding misaligned transits. The high nearby-source FPP is reconciled: injection-calibrated difference images exclude every catalogued neighbour that TRICERATOPS weighted, and this holds with a deliberately broadened PRF.

TIC 262843927. Sectors 31 and 88 failed because the star was off the CCD's science pixels (CCD columns about 2112 and 31, outside 45-2092). Nothing there can be salvaged, and nothing there is a non-detection. The transit shape is central ($b \approx 0.2$-0.4) at the catalogue density. Undiluted brown-dwarf or stellar occultors, and equal twins at 2P, are disfavoured by $\Delta\chi^2 \approx 10.6$-$11.3$ and would need $e \approx 0.9$. Gaia's RV constancy over 23 epochs limits a companion to the target to $M \sin i < 20$-$28\ M_J$. The reported NFPP = 0 holds only for small apertures. In a 3-pixel aperture two wide neighbours could supply the depth, so the protection comes from the image test, not the flux budget. Two findings weigh against it. The star sits about 0.2 mag above the single-star sequence for its colour, and it also shows Gaia astrometric excess noise (15.2σ) and RUWE 1.31, so an unresolved companion carrying roughly 15-20% of the light is likely. That would raise the radius about 10% if the transit is on the primary, or move it to the companion. Its 24.11-d signal also ranked only ninth in its own search (stitch S/N 8.86), although Stage C's blind search ranked it first.

TIC 193718989. The 9.8″ neighbour would need an 83-90% eclipse with the actual PRF throughput, beyond what a main-sequence pair can produce. ZTF DR24 detects it at normal brightness inside the predicted eclipse window at both period parities, and the calibrated image test disfavours it. The “4.91 versus 2.31” statistic was again a drift extrapolation driven by one partly covered 2020 event. The shape is planet-like but only mildly disfavours eccentric grazing brown dwarfs ($\Delta\chi^2$ 2.4-4.9), and Gaia RVs allow companions up to 70-124 M_J.

Luminosity for colour. Against 71,717 Gaia stars within 100 pc, only TIC 262843927 is over-luminous for its colour (about 0.2 mag). TIC 270573718 and TIC 193718989 sit on or slightly below the single-star sequence. The test cannot see companions fainter than about 2-2.5 mag (details).

Across all three. Public archival work has reached diminishing returns. The remaining alternatives (unresolved companions inside about 1-2″, diluted or eccentric eclipsing systems that fit the shape, and for 270 the timing question) need new observations. The smallest useful ones are named in the synthesis.

Inputs, provenance and the frozen protocol

Both prior reports were read in full: *SEVEN CANDIDATES / CONTINUATION AUDIT* (26 pp., Codex bundle `continue-the-seven-candidate-tess-analysis`) and *THREE TESS CANDIDATES / Further archival constraints* (22 pp.). The 22-page report was not attached to the request; it was found after the protocol freeze in `.claude\tess_three_audit`. It concerns TIC 382098683, 188830138 and 201247574, the current top tier, and supplied the QLP and TGLC product caveats and the TRICERATOPS source audit used here. The original batch-1 deep-vetting report (19 pp.) and its bundle were also read. The prior numbers quoted in the request were verified against those bundles. Every value checked reproduces, including the 0.29 / 8.19, 4.91 / 2.31 and 9.09 / 4.11 retrospective statistics, which were recomputed with the continuation's own code. The request described one point imprecisely: the TIC 270573718 “aperture” series is the survey's Stage-D TESScut 1-pixel aperture in eight sectors, not a freshly extracted series.

Discovery code: repository commit `df39814a9936` (branch `s1/survey`), preset `survey_k1`: periods 0.5-50 d, durations 0.02-0.35 d in 14 log steps, sectors ≤ 98 at native cadence, at least 3 transits, an 8,100-s search budget, then pre-gate, Stage A, S (bound_alias), B, C and D. All inputs were opened read-only and hashed (`manifests/input_manifest.csv`; 41 groups). The analysis protocol was frozen at 2026-10-18T21:08:37Z (SHA-256 `dd9f219ff2ecec63…`) before any post-discovery epoch was looked for. Four amendments are recorded in the file: A1 (report found; TGLC and free-shape timing added), A2 (post-hoc: the empirical timing null, a neighbour-selection fix and an RV-statistic fix), A3 (a bounded free-shape timing variant, a read-only tabulation of the discovery funnel, and the replay's summary statistics, fixed before any injected search had run) and A4 (the colour-magnitude over-luminosity check, added at the reader's request).

Times are BTJD = $\mathrm{BJD_{TDB}} - 2457000$ throughout; ZTF times were converted from $\mathrm{MJD_{UTC}}$ to $\mathrm{BJD_{TDB}}$ at Palomar. The packet fit stores marginal errors at cycle 0 only (none for TIC 262843927, whose central-transit covariance is singular). The “frozen” column below is a conditional linear ephemeris refit to per-event times from the discovery products, with errors inflated by $\sqrt{\chi^2}_{\mathrm{red}}$, and fixed before any later data were searched.

Target	packet P (d)	packet T0 (BTJD)	T14 (h)	FU3 frozen linear ephemeris	discovery products (sectors)
TIC 270573718	41.383701 ± 0.000101	1334.90375 ± 0.00379	3.76	41.383924 ± 0.000247; T(n=31) = 2617.80187 ± 7.4 min (×2.66)	TESS-SPOC 1800 s: 1,4,5,6,7,11; TESS-SPOC 600 s: 27,28,31,32,33,34,37; TESS-SPOC 200 s: 61,64,65,67,68; QLP 200 s: 87,88,94,95,98
TIC 262843927	24.108929 (no formal error: singular fit)	1332.78771 (no formal error)	3.71	24.108507 ± 0.000213; T(n=79) = 3237.37427 ± 10.2 min (×3.12)	TESS-SPOC 1800 s: 1,2,3,4; TESS-SPOC 600 s: 28,29,30,34; TESS-SPOC 200 s: 61,64,68,69; QLP 200 s: 95,96,97
TIC 193718989	24.532886 ± 0.000112	2000.07548 ± 0.00467	3.35	24.532835 ± 0.000276; T(n=39) = 2956.86107 ± 8.3 min (×2.46)	TESS-SPOC 1800 s: 25,26; TESS-SPOC 600 s: 40,52,53; TESS-SPOC 200 s: 79,80

Public-archive census

659 receipted requests (MAST, TESScut, TCE bulk files, ExoFOP, VizieR, Gaia TAP and DataLink, IRSA ZTF, ASAS-SN, LAMOST, RAVE, ESO, KOA, NASA Exoplanet Archive, SIMBAD and web search). Full notes and receipts are in `results/census/`.

- **No new epochs.** MAST holds SPOC FFIs through S109, TICA through S107 and QLP through S104. No sector after 98 contains any of the three stars. TIC 193718989 next falls on silicon in S117-S119 (May-July 2027). TIC 270573718's S97 appears only in a TICA footprint and is off the SPOC footprint by 27 px. TIC 262843927's S31 and S88 are off the science pixels (pixel check below). The retrospective splits are therefore the only “recurrence” tests possible.
- **Other reductions of the same photons** (sensitivity tests only): QLP, TARS, TGLC (fetched by direct path for sectors 1-55, which the MAST index omits), GSFC-ELEANOR-LITE, T16, TASOC. TIC 262843927 also has SPOC 2-min data in S1-S3.
- **Catalogues.** No SPOC TCE in 57 single- and multi-sector files (only TIC 262843927 was ever a SPOC 2-min target, in S1-S3, where each sector holds at most two transits of low S/N). No TOI or CTOI within 2′. No match in seven TESS eclipsing-binary catalogues, ASAS-SN, ATLAS, ZTF periodic, WISE or CSS variables, or the Gaia DR3 eclipsing-binary and non-single-star tables. Absences are not treated as negative science.
- **Gaia DR3 flags.** TIC 262843927: RUWE 1.31 and astrometric excess-noise significance 15.2, a hint of an unresolved companion. TIC 193718989: the Discrete Source Classifier returns P(binary star) = 0.999999; its reliability for single M dwarfs is not established here, so it is recorded as a flag only. TIC

270573718 and 193718989 are flagged VARIABLE (SOLAR_LIKE). No Gaia source lies within 5″ of any target.

- **Spectroscopy.** TIC 262843927 has two RAVE velocities (7.5 ± 2.0 and 3.1 ± 3.5 km/s) consistent with Gaia (8.25 ± 0.51). TIC 193718989 has one LAMOST DR11 spectrum (dM0, about −7.4 ± 2.0 km/s) consistent with Gaia (−5.44 ± 1.07). No high-resolution imaging or contrast curve exists for any target.
- **Gaia epochs.** Gaia DR3 epoch photometry, the only pre-TESS time series (36 epochs for TIC 270573718, 46 for TIC 193718989, 2014-2017), has no epoch within ±3σ of a predicted transit; TIC 262843927 has no Gaia epoch photometry.
- **Ground time series.** ZTF DR24 (TIC 193718989 and eight neighbours within 60″; the 9.8″ star has 2081 clean detections). ZTF and ATLAS forced photometry need accounts and were not used. ASAS-SN light curves were unavailable (service down since 13 October 2026).
- **Access failures** (not negative results): every ExoFOP per-target endpoint returns the literal body “Array” with HTTP 200 (the bulk TOI/CTOI tables do download); the NASA Exoplanet Archive cone search was blocked by Cloudflare (403).
- **Product caveats.** The QLP S99-S104 timing error affects none of these stars. The QLP S74-S79 fix touches only TIC 193718989's S79 file, which is the corrected v02 and was not used by discovery.
- **Prior public analysis.** TIC 193718989 is a candidate in github.com/eumemic/exoplanets (added 2026-10-08): P = 24.5329228 d, R_p 2.58 $R_\oplus$, TRICERATOPS FPP 0.029 (0.011 with Gaia/MOLUSC constraints). It uses the same TESS-SPOC photons and is not independent confirmation, but it matters for novelty and attribution.

Methods common to all three targets

Events. An event is admitted by coverage only (at least half the in-transit interval sampled, at least 3 in-transit and 4 baseline bins on each side). It is never admitted by its fitted sign. Each admitted event gets a signed depth from a fixed-shape, exposure-integrated batman template with a local linear baseline (quadratic as a check). Errors are white out-of-transit scatter times a time-averaging β factor, and pooled errors are inflated by $\sqrt{\chi^2}_{\mathrm{red}}$.

Reductions of the same photons. The discovery input (biweight-flattened); the same TESS-SPOC/QLP flux with local baselines; pipeline SAP; the survey's Stage-D TESScut aperture; TGLC aperture and PSF; and nine predeclared FU3 TESScut extractions (radius 1.0, 1.5 and 2.0 px circles and a neighbour-excluding aperture, each with an annulus or a plane sky, plus PRF scene photometry, run

only when the scene residual is at most 10% of the target peak). These are sensitivity tests, never independent detections.

Calibration. Injections at shifted epochs with the real transits masked measure depth and timing errors per cadence group. Random null epochs measure how often a single epoch shows a dip as deep as the median real event. For timing, strictly periodic copies of the transit were injected at 24 phases and timed with the identical method. This showed heavy-tailed timing errors that a robust-scatter calibration underestimates.

Host tests. Per-pixel transit-decrement maps are pooled per sector, with empirical per-pixel noise from 48 off-transit control maps. The tested sources are the target, every Gaia source within 2.5 px or brighter than $G_{target} + 6$ within 4 px, every TRICERATOPS nearby-scenario host, and every neighbour the flux budget finds capable in a 1.5-3 px aperture. Each is scored by a non-negative PRF amplitude fit. The score is calibrated by injecting the dip into the real cubes on the target and on each neighbour, using both the mission PRF and a PRF broadened by a 0.35-px Gaussian.

Flux budget. Gaia G/BP/RP give TESS magnitudes (Stassun et al. 2019). Mission-PRF throughput in apertures of 1.5-3 px gives the eclipse depth each neighbour would need. A main-sequence pair removes at most about 50% of its light, and an unresolved contaminant must lie within $\Delta T = -2.5 \log_{10}(\text{depth}/f_{max})$.

Physical models. Profile-likelihood and MCMC shape fits use per-event baselines profiled analytically, with (k, b, ρ_{eff}), an optional dilution and optional per-season timing offsets. The minimum eccentricity follows from $\rho_{eff}/\rho_\star$ (Dawson & Johnson 2012). Gaia DR3 RV constancy statistics were converted to companion-mass limits with a relative Monte Carlo statistic. TRICERATOPS scenario weights come from the continuation's four existing runs and 27 prior reweightings; no new draws were made.

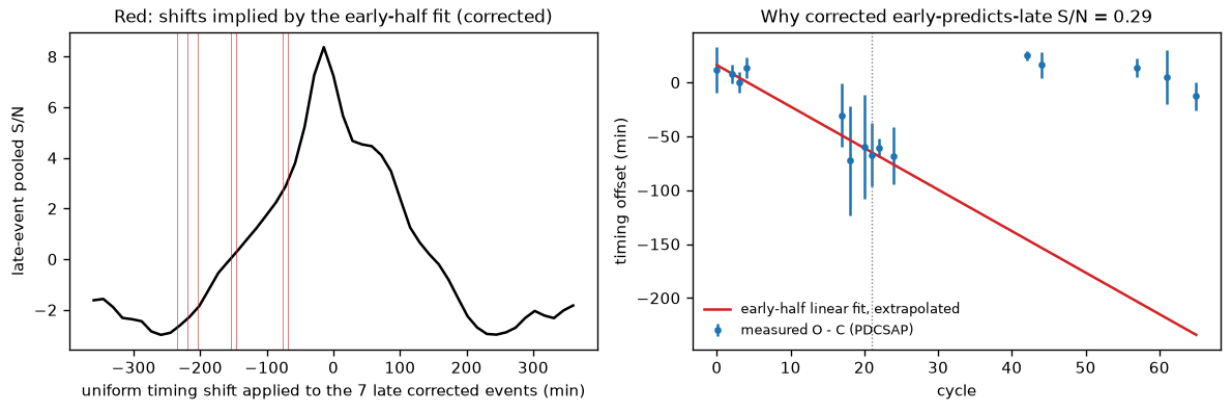
TIC 270573718 · P = 41.3837 d

Bottom line. A real, repeating dip in the discovery data on the named target, with all catalogued neighbours excluded as hosts by the calibrated image test. Its nature is unresolved. It is classified under protocol rule 5.7 as **(ii) persistent astrophysical dip, host and nature not resolved**. It is not (i), a processing artefact, because the dip is present in every target-centred raw-pixel extraction. It is not (iii), planet-favoured, because binary alternatives are disfavoured by $\Delta\chi^2$ of only 8.4 (PDCSAP) and 2.2 (pixel series), short of the predeclared 10.

(a) Initial claim and checked provenance

Packet fit: P = 41.383701 ± 0.000101 d, T0 = BTJD 1334.9038, T14 = 3.76 h, $R_\star$ = 0.463 $R_\odot$ (TIC; Gaia GSP-Phot gives 0.509). The discovery used 23 sectors: TESS-SPOC PDCSAP at 1800 s (S1-S11), 600 s (S27-S37) and 200 s (S61-S68), and QLP det_flux at 200 s (S87-S98). Of 16 predicted events, 15 are admitted. S95's event is fully quality-flagged in QLP. The QLP files are v01 and are not among the sectors with documented timing errors.

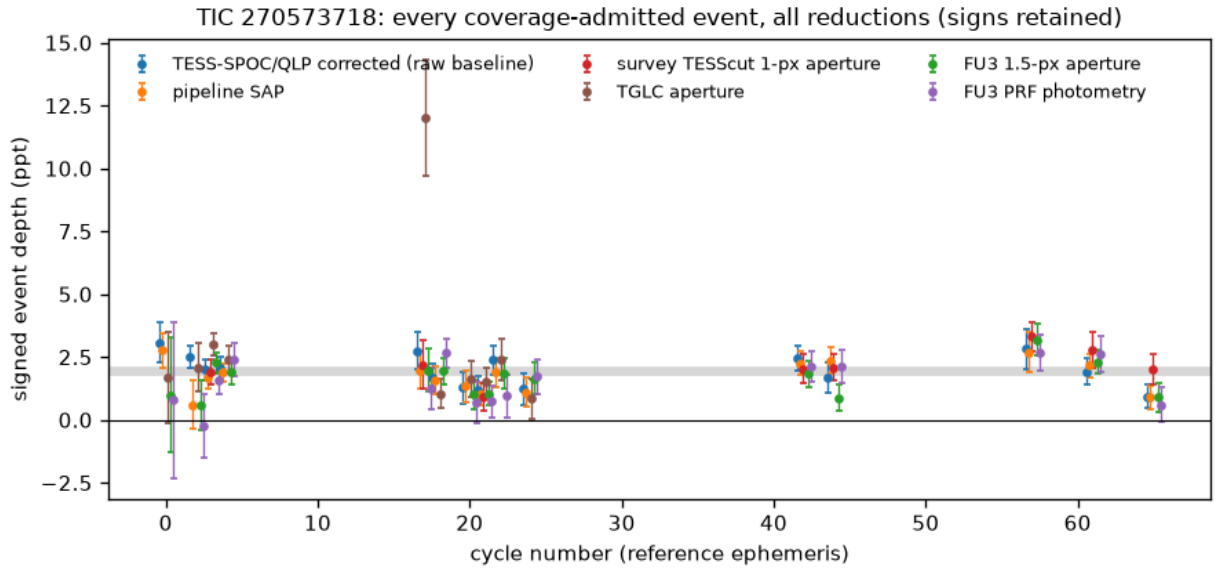
The contradiction, decomposed (`s01_270_reproduce.py`): the continuation's own code reproduces 0.2927 and 8.1926 exactly. Of the two series, only the corrected one has early events from 2018 (S1, S4, S5, S7) as well as 2020 (S27-S33). The aperture series' early half is S5, S27, S33 and S65. The corrected early fit finds a period correction of -3.85 min per cycle, robust to dropping any single early sector (S/N 0.27-0.74). Extrapolated to cycles 22-65, it shifts the predicted late transits by 1-4 h. On a common event set the corrected value rises to 3.63. With an epoch-only early fit it is 7.79, and at the reference ephemeris 7.25. The aperture series gives 8.24 at the reference ephemeris. The two reductions agree; the statistic does not, because it extrapolates a drift that the data constrain poorly.



Left: pooled S/N of the seven late corrected events versus a uniform timing shift; red lines mark the shifts implied by the early-half fit. Right: measured O–C (PDCSAP) and the early-half linear fit extrapolated forward.

(b) Tests executed, data sets and results

Reduction (same photons)	events	pooled depth (ppt)	S/N	p(constant depth)	positive
Discovery input (TESS-SPOC PDCSAP / QLP, biweight 0.75 d)	15	2.05 ± 0.17	12.0	0.130	15/15
Same product, local linear baselines, no biweight	15	1.94 ± 0.16	12.4	0.245	15/15
Pipeline SAP (no crowding correction)	15	1.72 ± 0.15	11.8	0.276	15/15
Survey Stage-D TESScut 1-px aperture	8	2.12 ± 0.25	8.3	0.173	8/8
TGLC aperture (sectors ≤ 55 only)	10	2.10 ± 0.40	5.2	0.000	10/10
TGLC PSF (sectors ≤ 55 only)	10	1.76 ± 0.40	4.4	0.002	9/10
FU3 TESScut 1.5-px aperture, annulus sky	15	1.72 ± 0.17	10.1	0.169	15/15
FU3 TESScut 1.5-px aperture, plane sky	15	1.61 ± 0.14	11.4	0.361	15/15
FU3 PRF scene photometry	15	1.72 ± 0.21	8.1	0.175	14/15



Every coverage-admitted event in six reductions, signs retained. Grey band: PDCSAP pooled depth ±1σ.

Timing. Per-event mid-times were measured in nine reductions. All of them show the same seasonal pattern relative to the reference ephemeris: 2018 ≈ +8 min, 2020-21 ≈ −50 to −64 min, 2023 ≈ +20-37 min, 2024-25 ≈ +3 to +13 min. The pipeline products, SAP, the survey TESScut aperture, TGLC aperture and PSF, and the FU3 pixel series all agree. A timestamp control rules out the FFI clock. WASP-62 b, in the same southern-CVZ cameras, has TESS-SPOC 600-s FFI transit times that match its SPOC 2-min times to within 0.16 to 0.60 min in each 2020-21 sector.

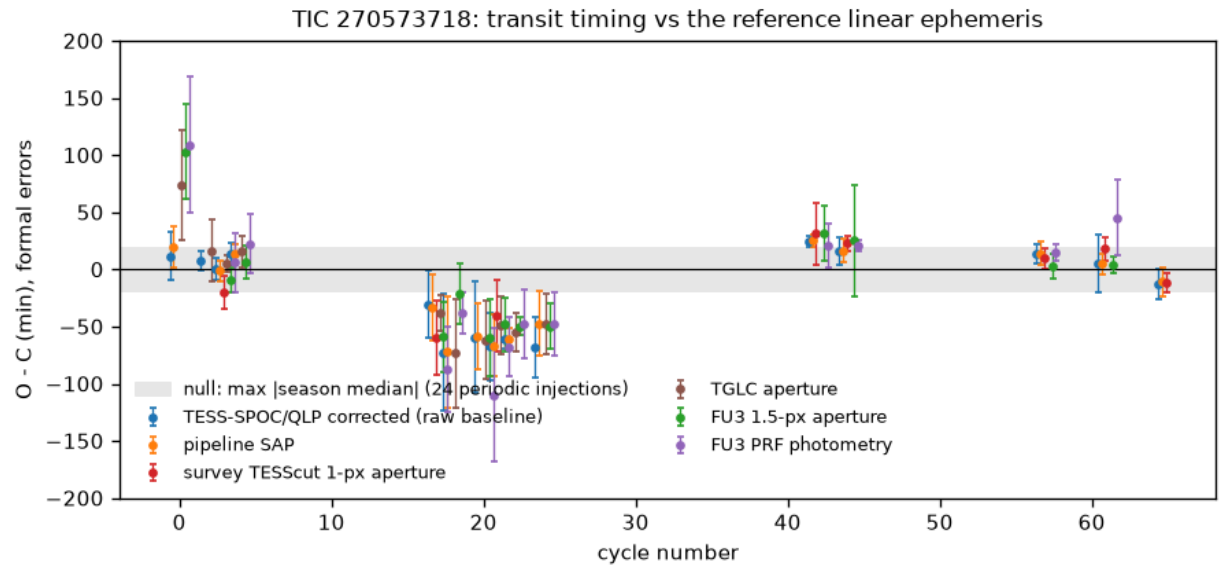
Significance depends on the error model. With formal errors, a linear ephemeris fails badly (χ^2 91.8/13). Injection-calibrated errors reduce this to 53.5 (adopted shape) or 25.4 (grazing shape). Strictly periodic injections into the same light

curve, timed identically, reveal heavy-tailed errors: the null season $\Delta\chi^2$ reaches up to 123, so the observed 86.2 has tail p = 0.08. Coherence is a different matter. The largest season-median offset in any null run is 18 min (PDCSAP), against 64 min observed with 6/6 events sharing the sign (tail p 0.04; similar in SAP and the pixel series). Under the predeclared rule (section 4: $\Delta\chi^2 \geq 11.8$ after inflation, the same pattern in an independent extraction, survival of leave-one-out), the timing variation is **not established**. The grazing-calibrated leave-one-out minimum is 9.5 once S34 is dropped.

Free-shape cross-check. Each event was refitted with a trapezoid whose depth, duration and ingress are free (amendment A1). Unbounded, the PDCSAP fit weakens the pattern (2020-21 season offset −28 min, event median −39 min, 4 of 6 events early). In the unclipped pixel series the unbounded fit latches onto outliers: the S1 fit is a 25-ppt, 1.1-h feature at the duration bound, so its +108-min 2018 offset is not interpreted. A post-hoc bounded variant (A3: depth 0-3× adopted, duration 0.6-1.6× adopted, isolated 5σ points removed from each window) gives the 2020-21 season about 50 min early with all six events sharing the sign in both series. The season pattern is therefore not an artefact of the fixed template. Its significance still rests on the injection null (p ≈ 0.04), which was computed for the template method only, and the 2023 and 2024-25 offsets differ between methods by up to 30 min.

Series	shape	season offsets 2018, 2020-21, 2023, 2024-25 (min)	2020-21 median, events sharing sign	season $\Delta\chi^2$ vs linear	median T14 (h) / ingress fraction	fits at a bound	S1 event fit
PDCSAP/QLP	unbounded	-2, -28, +18, +43	-39 min, 4/6	58.8	3.43 / 0.12		3.1 ppt over 6.9 h
PDCSAP/QLP	bounded	-3, -56, +52, +34	-51 min, 6/6	159.5	3.34 / 0.32	2	3.0 ppt over 6.0 h
FU3 pixel series	unbounded	+108, -57, +47, +33	-47 min, 6/6	1056.5	2.70 / 0.22		25.4 ppt over 1.1 h
FU3 pixel series	bounded	+21, -53, +48, +22	-47 min, 6/6	114.7	3.20 / 0.21	1	2.3 ppt over 5.0 h

Reduction	null injections	season $\Delta\chi^2$ (obs.)	tail p	max |season median O−C| (obs., min)	null maximum (min)	tail p
fu3pix_annulus_circ1p5	22	35.7	0.130	49.8	35.4	0.043
pdc_raw	24	86.2	0.080	63.9	18.2	0.040
sap_raw	24	84.5	0.120	59.5	27.4	0.040
tglc_aperture	9	-0.4	0.600	51.7	38.0	0.100



Transit times relative to the reference linear ephemeris in six reductions (formal errors). Grey band: the largest season-median offset produced by 24 strictly periodic injections into the PDCSAP light curve.

(c) Primary-signal reliability and selection limits

All 15 admitted events are positive in the pipeline products, and 14-15 of 15 in every FU3 pixel extraction. Pooled S/N is 8-12 and depths are consistent across events (p_{const} 0.02-0.47). Random null epochs reach the median event depth in 0.5% (PDCSAP) to 2.8% (SAP) of trials. No event lies within 0.8 d of a segment edge or near a momentum dump or scattered-light flag. These are all re-reductions of the discovery photons. Discovery selected the period and these events, and no later epoch exists. The selection replay is in the Selection replay section.

(d) Host attribution

The difference images tested 10 sources across 15 sectors. For every neighbour with a calibrated neighbour-hosted injection set, the observed summed $\Delta\chi^2$ lies above the 95th percentile of that set, under the mission PRF and the broadened PRF alike (8 of 8 excluded). Each observed value falls inside the target-hosted injection range. The 34.4″ G = 14.71 star needs only a 1% eclipse and carried most of the nearby-source FPP. Its observed score is 76.5, against a neighbour-hosted distribution centred near -382.

The earlier “conflict” between image tests and FPP is explained. TRICERATOPS computes NFPP from each neighbour's flux share in the aperture and never uses where the dimming sits in the image. Removing only the nearby scenarios whose hosts the image test excludes (TICs 270573715, 270573719, 270573720, 270573722) moves the FPP from 21.5%-45.7% to 3.5%-9.1%, and NFPP to 0.0007-0.0027. This is a conditional sensitivity bound, not a joint posterior. The image test and the light curve share photons, and the image test is calibrated only for the PRF family and perturbation tested. Unresolved companions within a few arcseconds behave like the target in these maps and remain open. Gaia RUWE is 1.12, and Gaia finds no source within 5″.

Tested source	sep. (arcsec)	eclipse needed	sectors	observed $\Sigma\Delta\chi^2$	target-hosted injections median [5-95 %]	neighbour-hosted injections median	neighbour-hosted, broadened PRF (p95)
Gaia DR3 4792202772323871104			1	4.6	10.7 [6.1, 29.6]	not injected	
Gaia DR3 4792203292016782592			6	39.1	36.1 [20.3, 61.1]	not injected	
Gaia DR3 4792202914059873280	35.3	0.01	15	76.5	99.5 [71.4, 131.0]	-381.8 (p95 -321.8)	-251.2
Gaia DR3 4792202948419610752	59.1	0.04	15	95.8	101.9 [74.2, 138.1]	-2394.1 (p95 -2154.5)	-1693.6
Gaia DR3 4792202737964129920	65.1	0.85	15	97.7	103.7 [77.1, 137.9]	-2281.2 (p95 -1925.6)	-1548.2
Gaia DR3 4792202776620920448	57.8	0.05	15	98.8	103.7 [76.2, 136.9]	-1858.8 (p95 -1675.2)	-1305.2
Gaia DR3 4792202944122578560	83.5	0.64	15	100.7	104.2 [76.5, 137.2]	-15883.2 (p95 -13846.5)	-10910.3
Gaia DR3 4792202536102541056	67.4	0.04	15	102.4	105.1 [78.0, 139.4]	-6478.4 (p95 -6060.7)	-4779.3
Gaia DR3 4792202016409623424	79.5	0.77	15	104.5	107.1 [79.1, 139.4]	-16582.3 (p95 -14694.0)	-11771.4
Gaia DR3 4792202845340395520	69.6	0.04	15	105.1	105.6 [77.2, 141.6]	-4905.5 (p95 -4399.6)	-3509.0

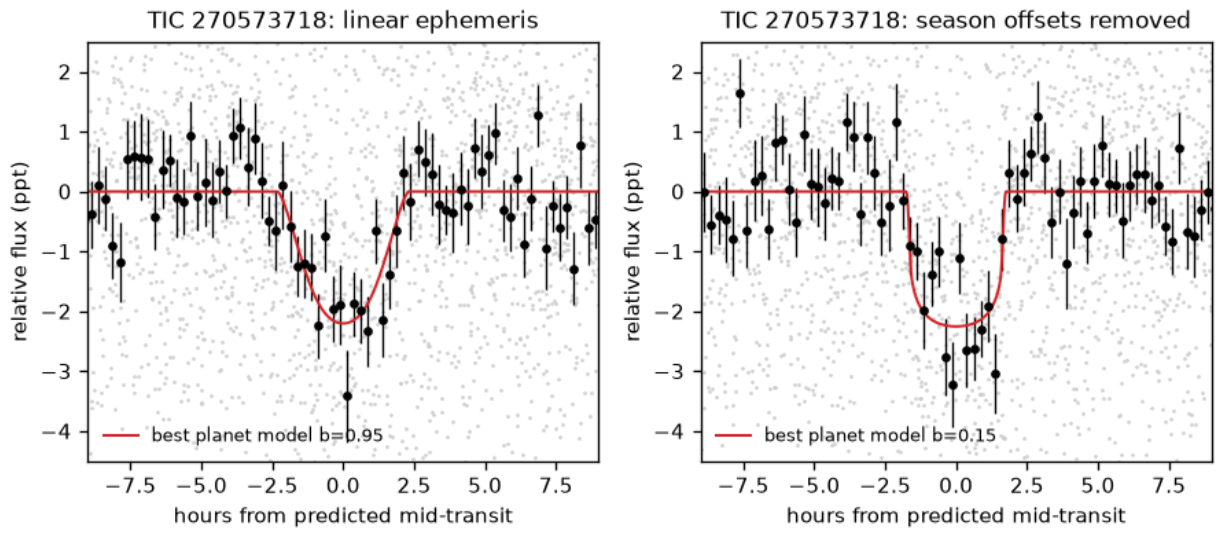
(e) Planet versus binary and brown-dwarf alternatives

Folded on a linear ephemeris, the transit looks V-shaped (best b = 0.95, ρ_{eff} 0.34 versus 6.52 g cm^{-3}). That shape is what allowed the earlier e ≈ 0.87 grazing companion and the 2P twins to fit as well as a planet. Freeing one timing offset

per season improves the fit by $\Delta\chi^2 = 50$ for four parameters and gives a flat-bottomed central transit: b = 0.15, $\rho_{eff} = 9.19$, catalogue density acceptable at $\Delta\chi^2$ 0.2. Large occultors (k ≥ 0.3 diluted, or undiluted 0.08-0.20 $R_\odot$ bodies needing e ≥ 0.81) are then disfavoured by $\Delta\chi^2$ 8.4 in PDCSAP but only 2.2 in the noisier pixel series. The shape argument therefore rests on the timing offsets being real, and is not independent evidence for them.

Odd and even depths agree within 1.2σ in all nine reductions, which says nothing against equal twins. A 0.52 ± 0.19 ppt dip at phase 0.5 appears in PDCSAP (phase-shift null p = 0.003). It falls to 0.35 in SAP and to 0.02-0.26 in the FU3 pixel series (p ≥ 0.07). It is not a robust secondary eclipse. If it were real it would require an eclipsing star, because no 2 $R_\oplus$ planet at 41 d produces a 0.5 ppt occultation. Gaia publishes no RV-constancy test at this brightness, so there is no RV limit.

Fit (PDCSAP/QLP unless noted)	k	b	ρ_{eff} (g cm^{-3})	e_{min}	$\Delta\chi^2$ at catalogue ρ	$\Delta\chi^2$ k ≥ 0.3 (diluted)	$\Delta\chi^2$ undiluted 0.08-0.20 $R_\odot$ occultor
linear ephemeris	0.0551	0.95	0.34 (cat. 6.52)	0.76	1.3	0.4	0.4 (needs e ≥ 0.84)
per-season timing offsets free	0.0438	0.15	9.19 (cat. 6.52)	0.11	0.2	8.4	8.4 (needs e ≥ 0.81)
FU3 pixel series, linear	0.0530	0.95	0.34		2.9	0.2	0.2
FU3 pixel series, season offsets	0.0518	0.91	0.83		0.6	2.2	2.2



PDCSAP/QLP events folded on the reference linear ephemeris (left) and after removing the fitted per-season timing offsets (right). Black points: 15-min bins.

(f) Model and prior sensitivity

Conditional on the named target hosting an undiluted transit, with the season offsets free: R_p = 2.31 [2.05, 2.60] $R_\oplus$ (68 %; density free), 2.25 [2.01, 2.50] $R_\oplus$ (circular orbit at the catalogue density); b = 0.57 [0.20, 0.82] / 0.46 [0.23, 0.63]; P(b > 0.8) = 0.20 / 0.02. Without the offsets the geometry is grazing-ambiguous and the radius is poorly bounded. TRICERATOPS (four runs, 27 prior reweightings each) gives package FPP 21.5%-45.7%, NFPP 0.187-0.438, target-host planet weight 54.3%-78.5% and any-host planet-label weight 61.5%-85.8%. Prior reweighting alone spans 1.1-98.6% FPP. None of these is a probability that a planet exists.

TRICERATOPS run	package FPP	NFPP	target-host planet (TP+PTP+DTP)	STP	any-host planet label	stellar-eclipse scenarios	image-conditioned FPP	image-conditioned NFPP	27 prior reweights (FPP)
seed_a	39.3%	0.3324	60.7%	5.5%	72.6%	27.4%	9.1%	0.0007	2.4% - 97.9%
seed_b	36.7%	0.3450	63.3%	1.9%	70.8%	29.2%	3.5%	0.0014	1.7% - 97.9%
noise_1p5	21.5%	0.1872	78.5%	2.6%	85.8%	14.2%	3.7%	0.0027	1.1% - 95.0%
native_bins	45.7%	0.4376	54.3%	1.6%	61.5%	38.5%	3.7%	0.0017	2.1% - 98.6%

(g) What changed since the prior reports

- The 0.29 / 8.19 contradiction is resolved: a drift extrapolation plus different event sets, not inconsistent reductions.
- New: a season-coherent timing structure in nine reductions, not produced by strictly periodic injections and not a clock error, but short of the claim rule.
- New: the grazing / e ≈ 0.87 and 2P-twin shape solutions are largely produced by timing smearing. With the offsets removed, the shape is central and planet-like.
- New: calibrated difference images exclude every catalogued neighbour behind the 19-44% NFPP.
- Unchanged: no new epoch exists, no RV constraint, and the half-phase dip is not robust.

(h) Viable, disfavoured and excluded-only-under-assumptions

- **Viable:** a sub-Neptune on the target (R_p ≈ 2.2-2.3 $R_\oplus$), possibly with transit-timing variations from a non-transiting companion; a planet on an unresolved bound companion (STP); a diluted eclipsing pair within a few arcseconds with a small radius ratio; an eclipsing pair in a hierarchical triple whose timing variations match the pattern.

- **Disfavoured:** an undiluted grazing brown dwarf or low-mass star on the target needing e ≥ 0.8, and equal twins at 82.77 d, but only if the season offsets are real ($\Delta\chi^2$ 2-8).
- **Excluded only under stated assumptions:** eclipses on any catalogued neighbour (mission and broadened PRF, real-cube injections); a light-travel-time explanation of the timing pattern (a ±40-min amplitude over about 5 yr needs a companion of several solar masses).

(i) Further public-data work

Little remains. Re-reductions agree, and a TESS clock error is excluded. Further reprocessing of the same photons would not test the timing structure. The next informative data are a resolved ground or space transit at the predicted epoch with a window of at least ±2 h (linear-ephemeris 1σ ≈ 7 min near the reference epoch, larger with drift), which would measure the next O–C and the shape. RV monitoring would test stellar-mass companions.

TIC 262843927 · P = 24.1089 d

Bottom line. The cleanest of the three. The dip is real in the discovery data, and every catalogued neighbour is excluded as host by the calibrated image test. The shape is a central transit at the catalogue density, and Gaia RVs exclude stellar and heavy brown-dwarf companions to the target. What remains is the generic unresolved-companion family (planet or eclipsing pair on a star within about 1-2″), which Gaia's astrometric excess noise leaves plausible. Recommendation: join the top follow-up tier.

(a) Initial claim and checked provenance

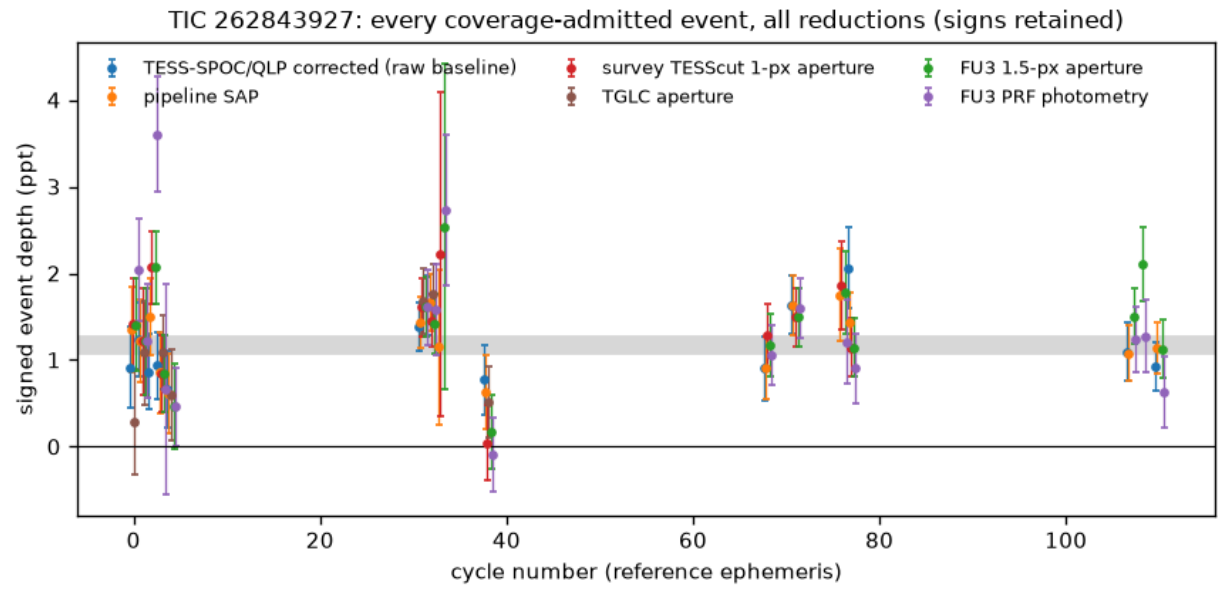
Packet fit: P = 24.108929 d, T0 = BTJD 1332.7877, T14 = 3.71 h, b ≈ 0. No formal P or T0 error exists because the covariance is singular at b = 0. The FU3 linear refit gives P = 24.108507 ± 0.000213 d. Discovery used 15 sectors: TESS-SPOC 1800 s (S1-S4), 600 s (S28-S34) and 200 s (S61-S69), and QLP 200 s (S95-S97). The “early-to-late 9.09 versus 4.11” difference is a noise difference, not a timing effect: at the reference ephemeris the late halves give 8.75 (corrected) and 4.27 (the survey's smaller, noisier TESScut aperture), and the early fits imply negligible drift.

Sectors 31 and 88 (`s09_262_s31_s88.py`). Through the cutout WCS the star falls on CCD column 2112.2 (S31, camera 4 CCD 2) and 31.3 (S88, camera 4 CCD 3). Both lie outside the science columns 45-2092, in the virtual overclock and underclock regions. The median image at the target position is 0.01 and 0.34 e^-/s, against about 845 e^-/s expected for a T = 12.0 star. tess-point agrees that the star was not on silicon, and no pipeline light curve exists for either sector. The prior

"gross failures" were geometric. No repair is possible, and these sectors are neither detections nor non-detections.

(b) Tests executed, data sets and results

Reduction (same photons)	events	pooled depth (ppt)	S/N	p(constant depth)	positive
Discovery input (TESS-SPOC PDCSAP / QLP, biweight 0.75 d)	14	1.21 ± 0.10	12.1	0.508	14/14
Same product, local linear baselines, no biweight	13	1.17 ± 0.10	11.2	0.393	13/13
Pipeline SAP (no crowding correction)	15	1.25 ± 0.10	12.6	0.687	15/15
Survey Stage-D TESScut 1-px aperture	12	1.34 ± 0.14	9.4	0.144	12/12
TGLC aperture (sectors ≤ 55 only)	7	1.15 ± 0.22	5.2	0.105	7/7
TGLC PSF (sectors ≤ 55 only)	7	1.36 ± 0.54	2.5	0.005	5/7
FU3 TESScut 1.5-px aperture, annulus sky	16	1.33 ± 0.12	11.0	0.120	16/16
FU3 TESScut 1.5-px aperture, plane sky	16	1.34 ± 0.13	10.1	0.022	16/16
FU3 PRF scene photometry	16	1.20 ± 0.18	6.7	0.002	15/16



Every coverage-admitted event in six reductions, signs retained. Grey band: PDCSAP pooled depth ±1σ.

Timing. Formal O−C scatter is large (linear χ^2 87.7/9), driven by single events, chiefly S61 at about −170 min in the pipeline products but −25 min in the survey TESScut aperture. The strictly periodic injection null reproduces this level (tail p 0.40; coherence tail 0.56), so there is no timing signal. The per-season offsets in the shape fit are within ±9 min.

Phase 0.387. The earlier nominal 8.9σ feature re-measures at 0.011 ± 0.092 ppt (PDCSAP) and between 0.01 and 0.15 ppt in all nine reductions. Individual

epochs are inconsistent with a constant depth. The earlier calibrated global p = 0.493 stands, and no secondary is claimed.

(c) Primary-signal reliability and selection limits

SPOC 2-min data (S1-S3; same epochs, different data path). 4 of 4 events are positive (1.13, 1.07, 1.00, 0.91 ppt), pooled 1.03 ± 0.19 ppt (p_{const} 0.98), consistent with the 1800-s FFI measurements of the same transits. This is a sensitivity check, not an independent epoch.

13-15 of 13-15 admitted events are positive in the pipeline products, and 15-16 of 16 in the FU3 pixel series. Pooled S/N is 10-12.6 (pipeline) and 6.7-11.8 (pixel series). Random null epochs reach the median event depth in 0.5%-6.1% of trials. The S1-S3 SPOC 2-min data are the same epochs and add no independent test. No later epoch exists.

(d) Host attribution

The reported NFPP = 0 came from TRICERATOPS's admission rule: no catalogued neighbour can make the 1.2 ppt dip in its aperture. That is true in small apertures. In the FU3 1.5-px aperture the nearest capable neighbour would need more than a 100% eclipse. In a 3-px aperture, two wide stars could supply it (65″, ΔT 3.9, about 9%; 103″, ΔT 1.9, about 20%). The calibrated image test settles this. All 5 tested sources score $\Sigma\Delta\chi^2$ +42 to +62 over 14 sectors, inside the target-hosted injection range. Every neighbour-hosted injection set lies far below, for both PRF versions (5 of 5 excluded).

The open family is unresolved companions. Gaia DR3 gives RUWE 1.31 and an astrometric excess-noise significance of 15.2, which is suggestive of a companion but not proof. TRICERATOPS's residual FPP is almost entirely STP (a planet on an unresolved bound companion). An unresolved eclipsing contaminant must have ΔT ≤ 7.19 (≤ 6.44 for a main-sequence pair). No contrast curve exists.

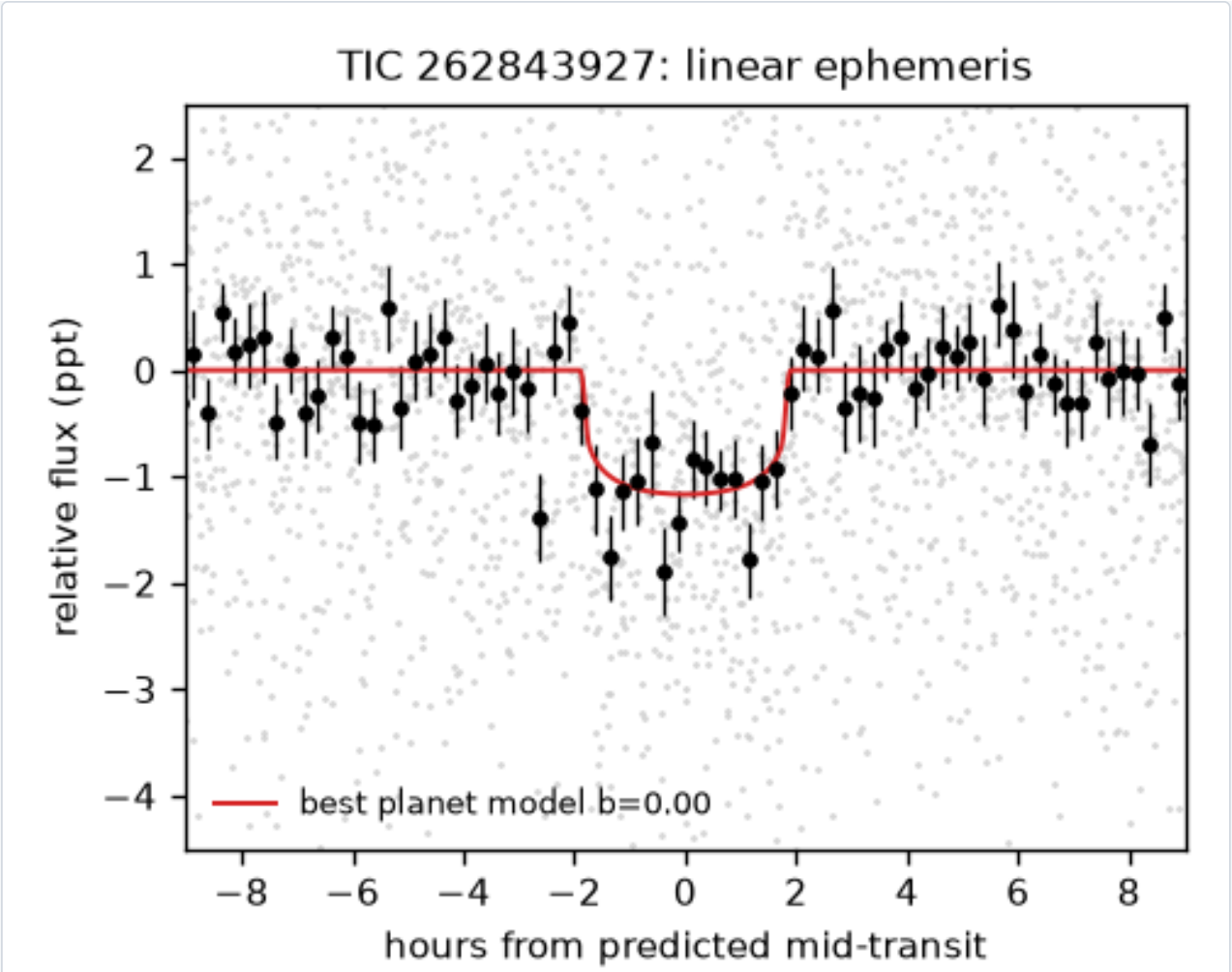
Tested source	sep. (arcsec)	eclipse needed	sectors	observed $\Sigma\Delta\chi^2$	target-hosted injections median [5-95 %]	neighbour-hosted injections median	neighbour-hosted, broadened PRF (p95)
Gaia DR3 4673750838475299968	46.2	0.65	14	41.8	51.9 [32.1, 74.4]	-548.1 (p95 -464.5)	-360.0
Gaia DR3 4673750911491494784	89.5	0.51	14	57.4	54.6 [31.1, 78.8]	-40836.9 (p95 -38718.0)	-29197.1
Gaia DR3 4673750911491491072	65.0	0.09	14	60.1	54.2 [33.3, 77.2]	-4630.7 (p95 -4148.2)	-3247.7
Gaia DR3 4673749983778555904	103.0	0.20	14	61.8	54.3 [33.3, 77.0]	-793871.9 (p95 -738690.0)	-601670.3
Gaia DR3 4673750048201317888	67.5	0.83	14	62.4	53.8 [32.8, 76.6]	-242.6 (p95 -160.9)	-115.6

(e) Planet versus binary and brown-dwarf alternatives

Shape: k = 0.0313, central (b 0.00-0.26), ρ_{eff} 4.23 versus 4.37 catalogue (e_{min} 0.01). Large occultors are disfavoured: diluted k ≥ 0.3 at $\Delta\chi^2$ 10.6, undiluted 0.08-0.20 $R_\odot$ bodies at 10.6-11.3, which would need e ≥ 0.89. Equal twins at 48.218 d imply k ≈ 1 and fall in the same disfavoured region. Odd and even depths agree within 1.9σ.

The phase-0.5 depth is -0.017 ± 0.093 ppt (PDCSAP). A circular eclipsing binary at P would therefore need a secondary below about 0.17 ppt, roughly 14% of the primary, which requires a companion much cooler than the eclipsed star. The earlier 2-ppt secondary-search sensitivity could not constrain diluted eclipses of this depth at all. Gaia RV constancy (23 transits, p = 0.977) limits a companion to the target, for the named target as host: M sin i < 20 M_J (circular, P), 22 (e = 0.8), 25-28 at 2P. This excludes stellar and heavy brown-dwarf occultors on the target. It says nothing about a blended pair, whose lines would not move the target's.

Fit (PDCSAP/QLP unless noted)	k	b	ρ_{eff} (g cm^{-3})	e_{min}	$\Delta\chi^2$ at catalogue ρ	$\Delta\chi^2$ k ≥ 0.3 (diluted)	$\Delta\chi^2$ undiluted 0.08-0.20 $R_\odot$ occultor
linear ephemeris	0.0313	0.00	4.23 (cat. 4.37)	0.01	0.2	10.6	10.6 (needs e ≥ 0.89)
per-season timing offsets free	0.0318	0.26	4.18 (cat. 4.37)	0.02	-0.0	11.3	11.3 (needs e ≥ 0.91)



PDCSAP events folded on the reference ephemeris. Black points: 15-min bins; red: best free-geometry model.

(f) Model and prior sensitivity

Conditional on the target hosting an undiluted transit: R_p = 1.98 [1.76, 2.22] $R_\oplus$ (68 %; density free), 1.94 [1.73, 2.15] $R_\oplus$ (circular orbit at the catalogue density); b = 0.42 [0.14, 0.70] / 0.24 [0.08, 0.46]; P(b > 0.8) = 0.08 / 0.00. TRICERATOPS: FPP 1.4%-3.9%, NFPP 0, target-host planet weight 96.1%-98.6%, any-host planet-label weight about 100%. Prior reweighting spans 0.15-28.7% FPP. These are conditional scenario weights, not a validation: the STP prior and Gaia completeness at small separations dominate them.

TRICERATOPS run	package FPP	NFPP	target-host planet (TP+PTP+DTP)	STP	any-host planet label	stellar-eclipse scenarios	image-conditioned FPP	image-conditioned NFPP	27 prior reweights (FPP)
seed_a	1.4%	0.0000	98.6%	1.4%	100.0%	0.0%	1.4%	0.0000	0.1% - 12.8%
seed_b	3.9%	0.0000	96.1%	3.9%	100.0%	0.0%	3.9%	0.0000	0.4% - 28.7%
noise_1p5	3.7%	0.0000	96.3%	3.7%	100.0%	0.0%	3.7%	0.0000	0.4% - 27.6%
native_bins	3.1%	0.0000	96.9%	3.0%	100.0%	0.0%	3.1%	0.0000	0.3% - 23.9%

(g) What changed since the prior reports

- The S31/S88 failures are explained as off-silicon geometry, not PRF, noise or crowding problems.
- NFPP = 0 is aperture-dependent. In a 3-px aperture two wide neighbours could supply the depth. The calibrated image test, not the flux rule, excludes them.
- New: Gaia RV mass limits on the target (M sin i < 20-28 M_J) and the large-occultor shape penalty with timing nuisance parameters.
- Radius now has a finite posterior: about 1.9-2.0 $R_\oplus$, consistent with the prior 1.82-2.41 profile range.
- The phase-0.387 feature stays unsupported, and the 9.09 / 4.11 split is explained by aperture noise.

(h) Viable, disfavoured and excluded-only-under-assumptions

- **Viable:** a 1.9-2.0 $R_\oplus$ planet on the target in a low-eccentricity orbit; a planet on an unresolved bound companion (STP, favoured by the astrometric noise if the companion is real); a diluted eclipsing pair with a small radius ratio on a star within about 1-2″.
- **Disfavoured:** grazing or eccentric brown dwarfs and low-mass stars on the target (shape $\Delta\chi^2 \approx 11$, RV); diluted pairs with radius ratio ≥ 0.3, including equal twins at 2P ($\Delta\chi^2 \approx 11$).
- **Excluded only under stated assumptions:** any catalogued neighbour as host (mission and broadened PRF injections); a companion to the target above about 20-28 M_J (Gaia RV error model).

(i) Further public-data work

Little remains. The SPOC 2-min S1-S3 light curves could refine the ingress shape slightly, but cover the same epochs. The decisive next data are high-resolution

imaging (to test the companion that Gaia's excess noise hints at) and a ground transit with resolved photometry, followed by RVs.

TIC 193718989 · P = 24.5329 d

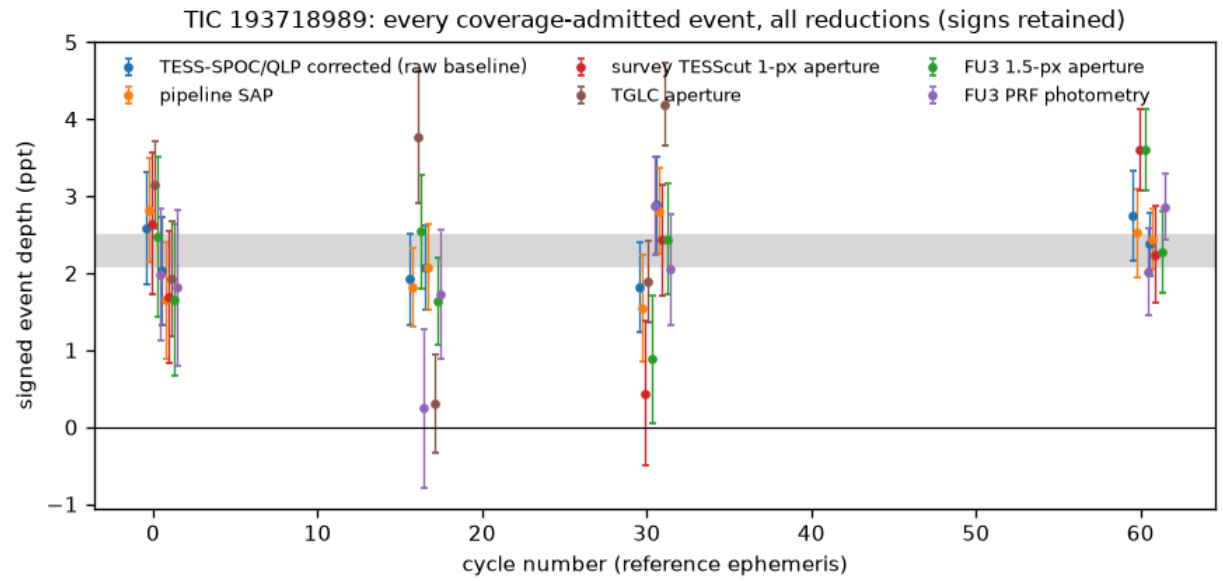
Bottom line. The archival ground data and the blend analysis improve host attribution materially. The 9.8″ neighbour is disfavoured by three lines: it would need an 83-90% eclipse, ZTF sees it at normal brightness inside the eclipse window, and the calibrated image test favours the target. The planet interpretation is weaker than for TIC 262843927, because the transit shape only mildly disfavours eccentric grazing brown dwarfs and Gaia RVs allow companions up to about 70-124 M_J. Recommendation: join the top tier, ranked below TIC 262843927. A resolved transit is no longer needed for the 9.8″ star, but is still the cleanest test of everything closer.

(a) Initial claim and checked provenance

Packet fit: P = 24.532886 ± 0.000112 d, T0 = BTJD 2000.0755, T14 = 3.35 h. Discovery used 7 sectors: TESS-SPOC 1800 s (S25, S26), 600 s (S40, S52, S53) and 200 s (S79, S80). All 8 events are admitted. The “4.91 versus 2.31” retrospective values reproduce exactly. Both are depressed by the early-half drift fit (2.16 min per cycle), which is driven by the partly covered S26 event; dropping S26 restores 9.35. At the reference ephemeris the late halves give 9.58 and 6.36. An independent public analysis (eumemic) finds the same signal in the same photons.

(b) Tests executed, data sets and results

Reduction (same photons)	events	pooled depth (ppt)	S/N	p(constant depth)	positive
Discovery input (TESS-SPOC PDCSAP / QLP, biweight 0.75 d)	8	2.35 ± 0.20	11.8	0.748	8/8
Same product, local linear baselines, no biweight	8	2.30 ± 0.20	11.4	0.879	8/8
Pipeline SAP (no crowding correction)	8	2.27 ± 0.20	11.6	0.696	8/8
Survey Stage-D TESScut 1-px aperture	6	2.50 ± 0.42	6.0	0.066	6/6
TGLC aperture (sectors ≤ 55 only)	6	2.56 ± 0.59	4.4	0.000	6/6
TGLC PSF (sectors ≤ 55 only)	6	2.33 ± 0.61	3.8	0.020	6/6
FU3 TESScut 1.5-px aperture, annulus sky	8	2.35 ± 0.30	7.8	0.136	8/8
FU3 TESScut 1.5-px aperture, plane sky	8	2.25 ± 0.29	7.8	0.071	8/8
FU3 PRF scene photometry	8	2.27 ± 0.25	9.1	0.355	8/8



Every coverage-admitted event in six reductions, signs retained. Grey band: PDCSAP pooled depth ±1σ.

Timing scatter in individual events (S26 −44 min, S52 +45 min) depends on the reduction. The FU3 1.5-px series is perfectly linear (χ^2 2.5/5), and the strictly periodic null reproduces the PDCSAP scatter (tail p 0.24). There is no timing signal.

(c) Primary-signal reliability and selection limits

8 of 8 events are positive in every pipeline product, and 7-8 of 8 in the pixel series. Pooled S/N is 11.4-11.8 (pipeline) and 5-10 (pixel series). Random null epochs reach the median event depth in 0-2% of trials. These are the discovery photons only; the next TESS epoch is in 2027.

(d) Host attribution: the 9.8″ neighbour and others

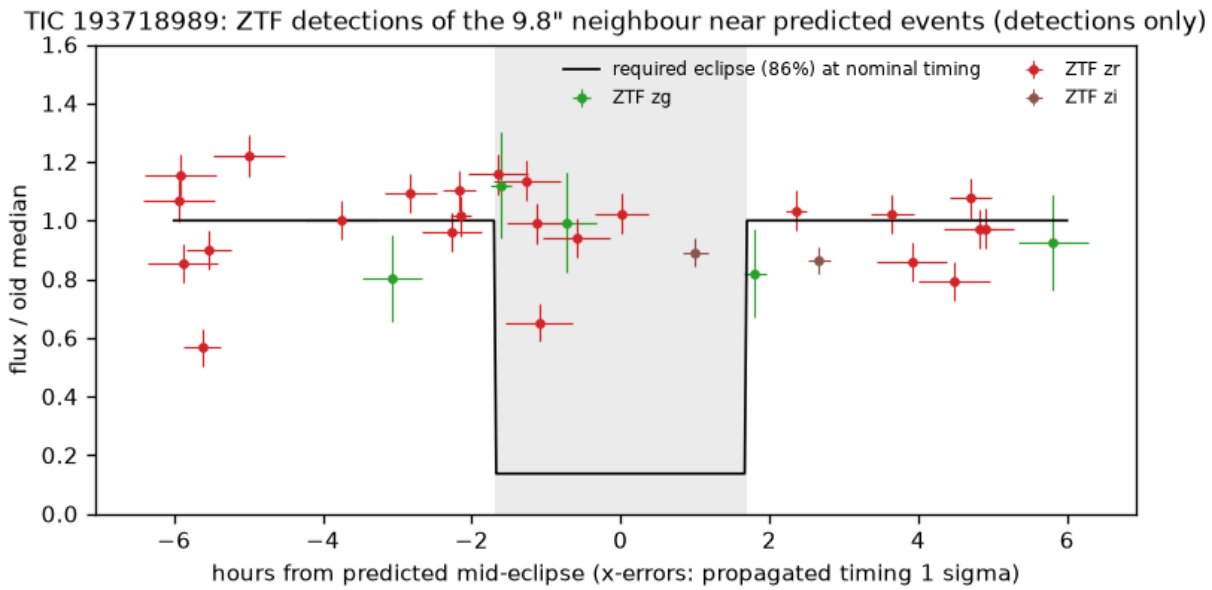
Flux budget. Gaia DR3 1346023547768414976 (G 19.59, BP−RP 1.85, 9.8″) has T ≈ 18.8, ΔT 6.4 from the target. With mission-PRF throughput it would need to lose 86% of its own light in the most favourable aperture and sector (0.83-0.90 across sectors with an RP-band flux ratio instead of T). That is beyond the roughly 50% a main-sequence pair can produce. The earlier 74% came from equal throughput and the RP band. TRICERATOPS assigned this star no weight, consistent with that physics.

ZTF DR24 (detections only; forced photometry is account-gated and was not used). The star has 2081 clean detections (zg 509, zr 1333, zi 239). Using the frozen ephemeris and its inflated covariance, propagated to each ZTF timestamp, 9 detections fall inside the nominal eclipse window, against 11.8 expected if the star never dims. There is no censoring deficit, although a deep eclipse would push it below the single-epoch limit. Two detections lie robustly inside the window even after a 3σ timing allowance. Both are even-parity cycles, at flux 0.935 ± 0.039 of normal, so any eclipse there is shallower than 13% (95%).

Odd-parity detections 0.6-1.1 h from predicted mid-eclipse are also at normal brightness, apart from one 0.46-mag-faint point next to a normal one in the same visit. Of the grid of eclipse models (depth 50-100%, durations 0.8-1.2 T14, ±3σ timing), 97% are contradicted by at least one detection.

Image test. The observed $\Sigma\Delta\chi^2$ for this star is 26.4. Target-hosted injections give a median of 19.7, and dips injected on the neighbour give a median of -30.5 (95th percentile -8.7; -6.2 with the broadened PRF). At 0.5 px separation the two hypotheses overlap more than for distant stars, so this line is the weakest of the three. The fitted single-source centroids scatter 2.5-9.3″ in directions unrelated to the neighbour (mean cosine about −0.05).

Across all 17 tested sources over 7 sectors, including the brighter stars at 65″ and 117″ that could supply the depth in large apertures, 12 of 12 with a neighbour-hosted injection set are excluded under both PRF versions. The rest are consistent with a target host. Remaining: unresolved companions within about 1-2″ (Gaia RUWE 1.04, no Gaia source within 5″, DSC binary flag recorded, no contrast curve).



ZTF DR24 detections of the 9.8″ neighbour within 6 h of predicted events, as flux relative to its own median. Horizontal bars: propagated 1σ timing. Black line: the eclipse it would need, at nominal timing. Detections only; non-detections are not available.

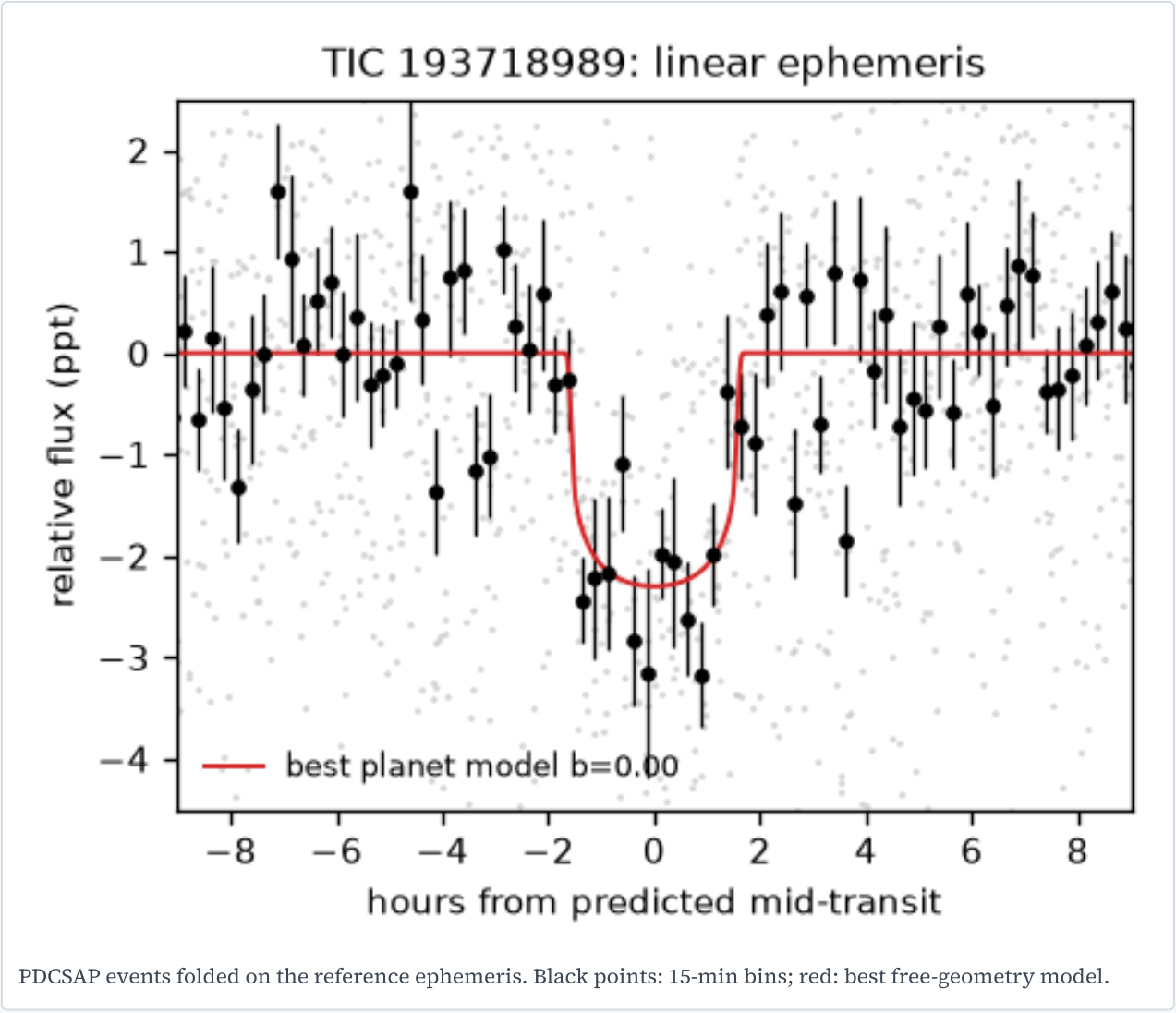
Tested source	sep. (arcsec)	eclipse needed	sectors	observed ΣΔχ²	target-hosted injections median [5-95 %]	neighbour-hosted injections median	neighbour-hosted, broadened PRF (p95)
Gaia DR3 1346023547768414976			7	26.4	19.7 [-49.9, 37.4]	-30.5 (p95 -8.7)	-6.2
Gaia DR3 1346023547770030080			7	49.1	42.0 [-56.9, 64.9]	not injected	
Gaia DR3 1346023543474451328			7	50.3	37.0 [6.2, 68.0]	not injected	
Gaia DR3 1346023371675752448			4	54.6	45.3 [27.7, 79.4]	-2146.6 (p95 -2037.1)	-1617.8
Gaia DR3 1346023440395239040			6	69.2	69.1 [42.9, 105.6]	not injected	
Gaia DR3 1346023612193934848			7	74.4	68.0 [6.8, 98.6]	not injected	
Gaia DR3 1346017706613825408	131.1	0.51	7	90.0	87.6 [64.2, 124.0]	-3671181.5 (p95 -3329230.0)	-2670304.3
Gaia DR3 1346023646553671936	45.7	0.24	7	91.4	81.1 [51.5, 121.8]	-386.8 (p95 -314.4)	-233.3
Gaia DR3 1346026571426323328			7	92.4	87.0 [62.4, 124.8]	-4208.9 (p95 -3254.5)	-2526.4
Gaia DR3 1346023509114600064	65.5	0.02	7	92.5	85.1 [57.6, 125.1]	-1415.9 (p95 -1220.1)	-980.4
Gaia DR3 1346023341611219456			7	92.7	80.4 [53.3, 117.7]	not injected	
Gaia DR3 1346026537066887808	78.8	0.31	7	96.0	85.1 [61.3, 124.5]	-23913.6 (p95 -21503.1)	-16892.7
Gaia DR3 1346023307251174656	116.6	0.05	7	96.0	87.1 [61.1, 125.8]	-206692.7 (p95 -180215.4)	-168352.6
Gaia DR3 1346023307251175552	104.2	0.55	7	96.2	85.7 [35.7, 124.1]	-9735.6 (p95 -5822.7)	-5589.9
Gaia DR3 1346023341610913280	71.3	0.26	7	96.6	85.1 [57.5, 124.0]	-5409.7 (p95 -4679.5)	-3567.1
Gaia DR3 1346023272891435264			7	97.3	89.0 [63.9, 128.9]	-1946.2 (p95 -1244.2)	-974.0
Gaia DR3 1346026601491052032	120.9	0.30	7	98.0	87.1 [61.7, 124.5]	-2347744.5 (p95 -2061293.9)	-1859117.9

(e) Planet versus binary and brown-dwarf alternatives

Shape: k = 0.0439, b 0.00-0.10, ρ_{eff} 6.61 versus 4.89 catalogue (e_{min} 0.10). Undiluted 0.08-0.20 $R_\odot$ occultors are disfavoured only by $\Delta\chi^2$ 2.4-4.9 and would need e ≥ 0.83. Diluted k ≥ 0.3 pairs, including equal twins at 49.066 d, are disfavoured by 2.4-4.9. Odd and even depths agree within 1σ. The phase-0.5 depth is 0.39 ± 0.26 ppt (null p 0.06), not significant and not reproduced in the PRF series. Gaia RV constancy (21 transits, p = 0.18) limits a companion to the target to M sin i < 70 M_J (circular), 96 (e = 0.8), 90-124 at 2P. This excludes most stars but not brown dwarfs.

Weighing population priors: the surviving target-host brown-dwarf solutions need e ≈ 0.83 and a grazing chord. Brown dwarfs around M dwarfs at 25-50 d are rare (the “desert”), and the grazing geometry occupies a small fraction of transiting orientations. These solutions exist geometrically but carry low prior weight. Two caveats: this is not a calibrated probability, and the Gaia RV limit does not reach them.

Fit (PDCSAP/QLP unless noted)	k	b	ρ_{eff} (g cm^{-3})	e_{min}	$\Delta\chi^2$ at catalogue ρ	$\Delta\chi^2$ k ≥ 0.3 (diluted)	$\Delta\chi^2$ undiluted 0.08-0.20 $R_\odot$ occultor
linear ephemeris	0.0439	0.00	6.61 (cat. 4.89)	0.10	0.0	2.4	2.4 (needs e ≥ 0.83)
per-season timing offsets free	0.0444	0.10	6.77 (cat. 4.89)	0.11	0.2	4.9	4.9 (needs e ≥ 0.83)



PDCSAP events folded on the reference ephemeris. Black points: 15-min bins; red: best free-geometry model.

(f) Model and prior sensitivity

Conditional on an undiluted transit of the target: R_p = 2.70 [2.38, 3.12] $R_\oplus$ (68 %; density free), 2.60 [2.32, 2.89] $R_\oplus$ (circular orbit at the catalogue density); b = 0.51 [0.16, 0.85] / 0.39 [0.14, 0.57]; P(b > 0.8) = 0.22 / 0.00. TRICERATOPS: FPP 1.7%-4.8%, NFPP 0.0009-0.0047, target-host planet weight 95.2%-98.3%. Prior reweighting spans 0.17-42.3% FPP. The residual FPP is almost entirely STP.

TRICERATOPS run	package FPP	NFPP	target-host planet (TP+PTP+DTP)	STP	any-host planet label	stellar-eclipse scenarios	image-conditioned FPP	image-conditioned NFPP	27 prior reweights (FPP)
seed_a	4.8%	0.0010	95.2%	4.7%	99.9%	0.1%	4.7%	0.0000	0.5% - 37.5%
seed_b	1.7%	0.0009	98.3%	1.6%	99.9%	0.1%	1.6%	0.0000	0.2% - 20.5%
noise_1p5	2.8%	0.0047	97.2%	2.3%	99.5%	0.5%	2.4%	0.0000	0.2% - 42.3%
native_bins	1.8%	0.0009	98.2%	1.7%	99.9%	0.1%	1.7%	0.0000	0.2% - 21.0%

(g) What changed since the prior reports

- The 9.8″ neighbour moves from “needs about 74%, ZTF limit from three nights” to disfavoured by three largely independent lines (main-sequence depth ceiling, timing-propagated ZTF detections at both parities, calibrated image injections).
- The 4.91 / 2.31 values are explained by one event's timing and the drift extrapolation.
- New: Gaia RV mass limits, finite radius posterior (about 2.6-2.7 $R_\oplus$), and the brightness limit for an unresolved contaminant.
- Unchanged: no independent epoch, and eccentric grazing brown dwarfs survive at low prior weight.

(h) Viable, disfavoured and excluded-only-under-assumptions

- **Viable:** a 2.6-2.7 $R_\oplus$ planet on the target; a planet on an unresolved bound companion (STP); a diluted small-radius-ratio pair within about 1-2″; an eccentric ($e \approx 0.83$) grazing brown dwarf on the target.
- **Disfavoured:** equal twins at 2P and other diluted pairs with radius ratio ≥ 0.3 ($\Delta\chi^2$ 2.4-5).
- **Excluded only under stated assumptions:** the 9.8″ neighbour (main-sequence pair, ZTF timing propagation, PRF injections); other catalogued neighbours (image injections); stellar-mass companions to the target above about 70-90 M_J in circular orbits (Gaia RV error model).

(i) Further public-data work

Little remains. ZTF forced photometry would extend the neighbour test to non-detections, but needs an account. Better is a resolved multiband transit, which measures the target's depth directly (the neighbour is 9.8″ away), followed by RVs reaching the brown-dwarf regime (K ≈ 1-7 km/s) and high-resolution imaging.

Discovery selection: the funnel and a bounded replay

The funnel that chose these signals. S1 batch 1 searched 2,000 stars. 1,476 stars produced at least one pre-gate candidate (4,655 in all). Stage A passed 141 of them on 77 stars (25 stars had two or more), Stages S and B passed 127 and 127, and Stage C's blind search passed 32, one per star. Review then kept 23 SUBMIT/HOLD stars, the earlier continuation took seven, and these three come from those seven. All three passed every stage. The table lists each pre-gate candidate on the three stars.

The unfavourable fact is TIC 262843927. Its 24.11-d signal ranked only ninth in its own search (stitch S/N 8.86, the lowest of the three), behind eight others on the same star, including several stronger but spurious ones. Two other periods on that star (14.52 d and 30.23 d) also passed Stages A, S and B and failed only at Stage C, because the blind TLS's strongest peak was at 106.67 d. For the 24.11-d candidate the blind TLS's strongest peak was the candidate itself (SDE 25.9). The light curve of this star is busy, and the signal was found near the search threshold. TIC 270573718 and TIC 193718989 were their stars' top-ranked signals.

Star	search rank	period (d)	stitch S/N	transits	discovery funnel
TIC 270573718	1	41.3839	9.91	15	A PASS → S PASS → B PASS → C PASS
TIC 270573718	2	20.6929	9.45	30	A FAIL
TIC 262843927	1	41.7698	15.00	4	A FAIL
TIC 262843927	2	40.3429	19.99	8	A FAIL
TIC 262843927	4	14.5169	11.48	25	A PASS → S PASS → B PASS → C FAIL
TIC 262843927	5	45.0711	11.30	4	A FAIL
TIC 262843927	6	49.2047	10.90	3	A FAIL
TIC 262843927	7	30.2341	10.51	12	A PASS → S PASS → B PASS → C FAIL
TIC 262843927	9	24.1089	8.86	10	A PASS → S PASS → B PASS → C PASS
TIC 262843927	10	19.7956	12.46	15	A FAIL
TIC 262843927	14	15.9265	9.41	22	A FAIL
TIC 193718989	1	24.5335	10.40	8	A PASS → S PASS → B PASS → C PASS
TIC 193718989	2	23.6748	7.33	8	A FAIL
TIC 193718989	3	40.6208	10.02	4	A FAIL
TIC 193718989	5	12.2449	9.36	14	A FAIL
TIC 193718989	6	20.5672	7.41	3	A FAIL

The frozen search (`survey.run_job`, preset survey_k1, 8,100-s budget, identical pre-gate) and Stage A (`funnel_k2.py`, S mode bound_alias, C variant V3) were run in an isolated data root on fixed sets, chosen before any flux was read. The sets are the three targets' light curves with the sign inverted (dips become bumps), 24 inverted matched controls from batch 1 (same slice, $|\Delta Tmag| \leq 0.5$, usable sectors within ±30%), and 24 other matched controls with each candidate's own transit injected by the frozen injection code at a random epoch. Limits: 24 CPU-h and 4 workers, with no extension. Finished 2026-10-19T05:57:50Z: 51 of 51 searches, 17.2 CPU-h used.

What this covers: the search's look-elsewhere space over 0.5-50 d, its grids, gaps and budget, and the pre-gate and Stage A cuts, on matched noise. What it does not cover: the S/B/C/D stages, the human/Codex review that picked seven of the 23 SUBMIT/HOLD stars, and batch-to-batch choices. Inverted light curves are only a partial null, because asymmetric systematics (PDC-made dips) do not invert symmetrically. The survey's own K1 calibration on 250 uninjected stars found 0.088 natural candidates per star surviving Stage S. Those are real-star false alarms (systematics, variables, binaries), not pure noise.

Inverted (null) group	stars searched / planned	pre-gate passes	largest pre-gate stitch S/N	Stage A jobs	Stage A passes	stars with a pass: 95% upper limit	same stars, real data: pre-gate / Stage A passes
The three targets	3 / 3	5	24.3	5	**0**	< 63%	16 / 5
Controls matched to TIC 270573718	8 / 8	33	352.4	33	**0**	< 31%	23 / 0
Controls matched to TIC 262843927	8 / 8	15	13.0	15	**0**	< 31%	21 / 0
Controls matched to TIC 193718989	8 / 8	15	37.8	15	**0**	< 31%	9 / 0
All inverted stars	27 / 27	68	352.4	68	**0**	< 11%	

Injected signal	stars searched / planned	expected S/N (median)	found at P, P/2 or 2P (pre-gate)	passed Stage A (68% interval)	recovered stitch S/N, median [range]	real candidate's discovery stitch S/N
TIC 270573718 transit	8 / 8	12.1	6 / 8	**5 / 8** (39%-82%)	8.2 [7.3-12.5]	9.91 (rank 1)
TIC 262843927 transit	8 / 8	9.9	6 / 8	**6 / 8** (51%-91%)	8.1 [7.3-16.1]	8.86 (rank 9)
TIC 193718989 transit	8 / 8	14.0	6 / 8	**6 / 8** (51%-91%)	11.8 [8.3-14.1]	10.40 (rank 1)

Null result. The 27 inverted light curves searched gave 68 pre-gate candidates, with stitch S/N up to 352.4, well above the three real candidates (8.9-10.4). Search S/N alone therefore does not separate these signals from noise. Stage A does: none of the 68 inverted candidates passed it. With 27 stars and no passes, the per-star rate of noise-only Stage A passes is below 11% (95%). That bound is too weak to make the batch's survivors secure by itself: at the bound, the 1,476 batch-1 stars with a pre-gate candidate could yield up to about 35 noise signals reaching Stage C (using the observed 23% Stage A→C survival of real-data passes, an assumption for noise), against 32 actual Stage C survivors. The replay shows that Stage A is not easily fooled by the same noise; each signal's security still rests on its own per-event and per-reduction evidence.

Recovery. TIC 270573718's transit was recovered through Stage A on 5 of 8 matched stars (pre-gate match 6 of 8); recovered stitch S/N median 8.2 against the real candidate's 9.91 (1 of 6 recoveries at or above it). TIC 262843927's transit was recovered through Stage A on 6 of 8 matched stars (pre-gate match 6 of 8); recovered stitch S/N median 8.1 against the real candidate's 8.86 (2 of 6 recoveries at or above it). TIC 193718989's transit was recovered through Stage A on 6 of 8 matched stars (pre-gate match 6 of 8); recovered stitch S/N median 11.8 against the real candidate's 10.40 (4 of 6 recoveries at or above it). Recovery of 5-6 in 8 means the search finds signals like these most of the time but not always. Each real candidate's discovery S/N lies inside the range its injections reached, so the replay shows no sign that the measured depths are strongly inflated by selection. With 8 injections per candidate this is a coarse check.

Twice-period and long-period checks (targeted; not part of the survey's look-elsewhere space): Stage C's blind TLS (0.5-110 d) already ranked P first for all three, with 2P peaks lower (TIC 270573718: 82.77 d at SDE 40.3 versus 58.2 at P; TIC 193718989: 49.07 d at SDE 16.4 versus 37.9). The targeted box searches beyond the 50-day ceiling are below.

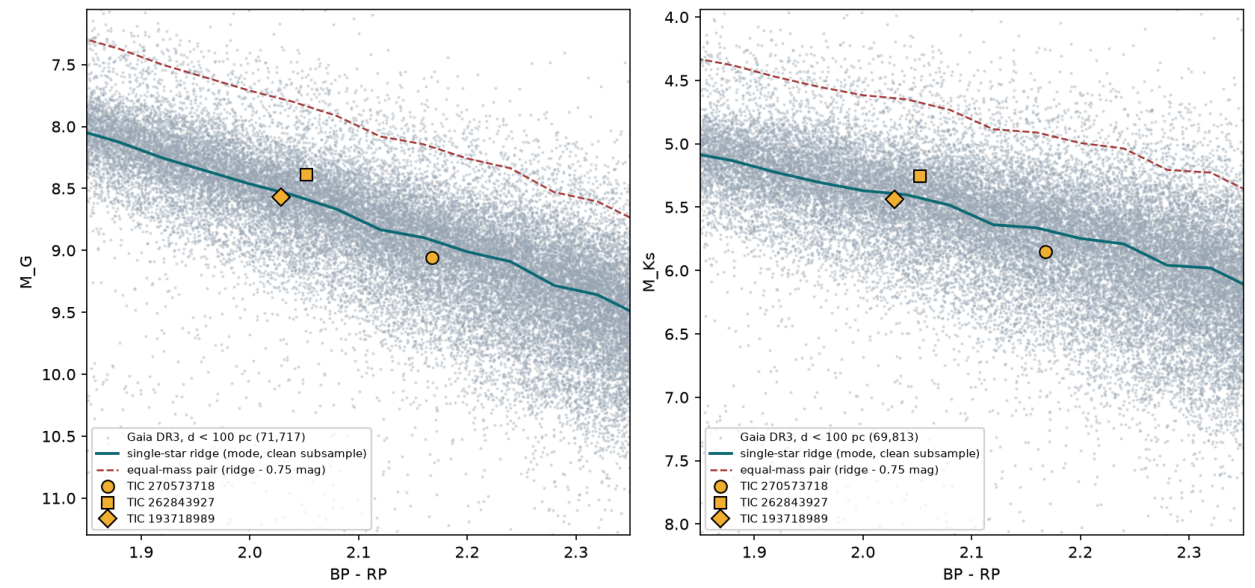
Target	series	band	top long-period peaks	top peaks after masking the candidate's transits
TIC 270573718	survey_detrended	60-130 d	82.767 d (2.000 P, SDE 11.3); 124.152 d (3.000 P, SDE 9.4); 62.076 d (1.500 P, SDE 7.6)	112.004 d (SDE 5.7, 3.19 ppt); 114.794 d (SDE 5.4, 2.99 ppt); 0.517 d (SDE 5.1, 0.14 ppt)
TIC 270573718	fu3pix_annulus_circ1p5	60-130 d	115.193 d (2.784 P, SDE 6.1); 96.136 d (2.323 P, SDE 5.8); 120.140 d (2.903 P, SDE 5.8)	115.314 d (SDE 6.9, 175.72 ppt); 119.766 d (SDE 6.6, 201.31 ppt); 96.135 d (SDE 6.6, 201.31 ppt)
TIC 262843927	survey_detrended	40-100 d	98.787 d (4.098 P, SDE 6.2); 70.547 d (2.926 P, SDE 6.2); 81.500 d (3.380 P, SDE 6.2)	113.735 d (SDE 7.3, 8.95 ppt); 110.389 d (SDE 7.3, 8.95 ppt); 100.510 d (SDE 7.3, 8.95 ppt)
TIC 262843927	fu3pix_annulus_circ1p5	40-100 d	99.260 d (4.117 P, SDE 6.5); 85.079 d (3.529 P, SDE 5.5); 72.193 d (2.994 P, SDE 5.5)	123.049 d (SDE 9.5, 120.22 ppt); 120.920 d (SDE 9.0, 124.61 ppt); 105.805 d (SDE 9.0, 124.61 ppt)
TIC 193718989	survey_detrended	40-100 d	49.067 d (2.000 P, SDE 7.2); 91.999 d (3.750 P, SDE 6.1); 61.332 d (2.500 P, SDE 6.1)	85.754 d (SDE 5.3, 2.20 ppt); 128.631 d (SDE 5.3, 2.20 ppt); 77.179 d (SDE 5.3, 2.20 ppt)
TIC 193718989	fu3pix_annulus_circ1p5	40-100 d	65.728 d (2.679 P, SDE 3.2); 88.663 d (3.614 P, SDE 3.2); 72.281 d (2.946 P, SDE 3.2)	120.500 d (SDE 4.3, 11.60 ppt); 65.658 d (SDE 4.3, 11.60 ppt); 88.585 d (SDE 4.3, 11.60 ppt)

Reading the table: in the discovery input, the long-band peaks are aliases of P (2P, 3P, 1.5P) or weak (SDE about 6). No long-period solution beats the reported period, and no 2P alias stands out where odd/even depths agree. After the candidate's transits are masked, no residual peak exceeds SDE about 7. In the FU3 pixel series (outliers not clipped), the residual peaks are driven by isolated outlier points (implied depths above 100 ppt) and are not interpreted.

Is each star over-luminous for its colour?

An unresolved companion of similar brightness lifts a star above the single-star main sequence at its colour. An equal-mass pair sits 0.75 mag above it, a companion 1.5 mag fainter about 0.24 mag above, and one 3 mag fainter only 0.07 mag above. This check was added at the reader's request (amendment A4). Each target was placed on four empirical colour-magnitude diagrams built from Gaia DR3 stars within 100 pc (71,717 after the BP/RP flux-excess cut), with 2MASS Ks from Gaia's own cross-match. The reference is the single-star ridge, the most common absolute magnitude at each colour among 44,312 clean stars (RUWE < 1.2, no Gaia binary solution, no multi-peak flag). Metallicity spreads single stars around that ridge, and the spread measured from the same clean stars is given below. Extinction is neglected at 85 pc.

Target	diagram	ΔM (mag; negative = brighter than the ridge)	clean stars at this colour at least as bright	clean-star 16-84% range	clean stars faint enough to be half of an equal-mass pair	Gaia RUWE / excess-noise significance
TIC 270573718	M_G vs BP−RP	+0.14	54%	-0.18 to +0.54	5%	1.12 / 0.8
	M_G vs G−RP	+0.11	55%	-0.21 to +0.52	5%	
	M_{Ks} vs BP−RP	+0.17	57%	-0.21 to +0.58	5%	
	M_{Ks} vs G−Ks	+0.25	57%	-0.21 to +0.73	9%	
TIC 262843927	M_G vs BP−RP	-0.20	11%	-0.16 to +0.42	10%	1.31 / 15.2
	M_G vs G−RP	-0.23	11%	-0.19 to +0.44	12%	
	M_{Ks} vs BP−RP	-0.17	15%	-0.17 to +0.47	11%	
	M_{Ks} vs G−Ks	-0.17	17%	-0.19 to +0.66	19%	
TIC 193718989	M_G vs BP−RP	+0.04	47%	-0.15 to +0.41	4%	1.04 / 0.0
	M_G vs G−RP	+0.04	48%	-0.18 to +0.42	4%	
	M_{Ks} vs BP−RP	+0.04	46%	-0.17 to +0.46	5%	
	M_{Ks} vs G−Ks	+0.01	38%	-0.19 to +0.66	12%	



The three targets against Gaia DR3 stars within 100 pc. Teal: single-star ridge (most common absolute magnitude of clean stars at each colour). Red dashed: where an equal-mass pair of ridge stars would sit.

TIC 262843927 is mildly over-luminous in all four diagrams (ΔM -0.17 to -0.23 mag). Only 11%-17% of clean stars at its colour are as bright. On its own that is inside the single-star spread, because a metal-rich single star can sit there. It does not stand alone. This is also the only target with an elevated RUWE (1.31) and a significant astrometric excess noise (15.2σ), and those measure the image's motion and shape, not its brightness. Together they make an unresolved companion the most economical reading, though not a proven one. If the primary sits on the ridge, the companion is about 1.5-1.9 mag fainter in G and carries 14%-19% of the light. An equal-mass pair is disfavoured but not excluded: 10%-19% of clean single stars at this colour are faint enough to make one look like this star. The consequences for the transit are as follows. If it is on the primary, the depth is diluted and the radius rises by about 8-11% (up to about 41% for an equal pair). If it is on the companion, the undiluted depth is about 6-8 ppt on an M2-M3 dwarf, a body of roughly 3-4 $R_\oplus$. The transit duration's agreement with the primary's density would then be a coincidence. The diluted-pair cases are already inside the dilution-free k-profiles. High-resolution imaging, already the recommended first observation for this star, tests the companion directly.

TIC 270573718 and TIC 193718989 are not over-luminous. TIC 270573718 sits slightly below the ridge (ΔM +0.11 to +0.25 mag) and TIC 193718989 on it (+0.01 to +0.04 mag). Both have RUWE near 1 and no astrometric excess noise. An equal-mass pair would need the primary to lie in the faintest 4%-12% of single stars at those colours (5% or less in the three diagrams with the narrower single-star spread). For TIC 193718989 this does not support Gaia's DSC “binary” probability (0.999999), which is derived from the low-resolution spectra rather than the colour-magnitude position.

Limit. The test cannot see companions more than about 2-2.5 mag fainter in G, because their lift (< 0.1-0.15 mag) is inside the single-star spread. Those faint companions are the ones that matter for most blended-eclipse scenarios, so a normal colour-magnitude position does not exclude them.

Cross-candidate synthesis

Target	repeating dip	target is host	planet-like body	main surviving alternatives	tier and smallest decisive observation
TIC 262843927	HIGH	MODERATE	MODERATE	Unresolved companion (excess noise 15.2σ, RUWE 1.31, about 0.2 mag over-luminous for its colour); planet on the companion; diluted small-k pair inside about 1-2″	Top tier. High-resolution imaging first (it tests the likely companion); resolved transit at 24.1089 d; RVs to about 1 km/s.
TIC 193718989	HIGH	MODERATE	LOW TO MODERATE	Eccentric grazing brown dwarf on the target (RV-allowed); STP; unresolved companion	Top tier, close behind 262. Resolved multiband transit; RVs to the brown-dwarf regime; imaging.
TIC 270573718	HIGH	MODERATE	LOW	Timing structure (TTV versus eclipsing pair in a triple versus noise); diluted pair; STP	Retain, second tier. A transit with a ≥±2 h window measures the next O−C and the shape; RVs.

The confidence labels are qualitative and conditional on the stated assumptions: Gaia completeness outside about 1-2″, the mission-PRF family with the tested perturbation, main-sequence physics for blended pairs, and the Gaia RV error model. They are not probabilities. “Repeating dip: HIGH” means present in 5-9 re-reductions of the discovery photons with low null false-dip rates. No independent epoch exists, and the replay is reported separately.

Should TIC 262843927 or TIC 193718989 join the top tier? Both should. TIC 262843927 has the strongest planet-versus-binary case of the three: a planet-like central shape at the catalogue density, and RVs that exclude stellar and heavy brown-dwarf companions to the target. Its host case is weaker than first stated. The star is about 0.2 mag over-luminous for its colour and has significant astrometric excess noise, so an unresolved companion is likely, and the transit could be on it. Its signal was also found near the search threshold. TIC 193718989 has the cleaner host case: no sign of a companion, and its 9.8″ neighbour is excluded (its host attribution is now about as good as 201247574's in the prior report). Its planet-versus-brown-dwarf separation is weaker. 262 keeps a narrow

lead because a companion would change its radius and host star, not its planetary character, whereas 193's brown-dwarf alternative would remove the planet. The order between the two is close.

TIC 270573718: retain, downgrade below both. It should not be dropped, because the dip is real in the discovery data, its host is the target among catalogued stars, and the earlier contradiction and NFPP conflict are both resolved. Its planet interpretation depends on whether the season timing structure is real. If it is, the shape favours a small central transiter and the system may be dynamically interesting. If it is not, the shape is grazing-ambiguous and the binary alternatives come back. Future epochs need wider windows than the linear ephemeris suggests.

Diminishing returns. For all three, further work on existing public data would re-reduce the same photons or re-weight the same scenarios. The remaining alternatives (unresolved companions within about 1-2″, small-radius-ratio diluted pairs, the 270 timing question, and brown dwarfs for 193) need new data. The single smallest observations: for 262, one high-resolution image (AO or speckle reaching Δmag about 6-7 at 0.1-2″); for 193, one resolved multiband transit; for 270, one transit with a wide window (it tests timing and shape together). No observing requests were made.

Test matrix keyed to the prior reports

Target	Test family	Prior status	FU3 action	FU3 outcome
TIC 193718989	Independent recurrence	RETROSPECTIVE_ONLY	Census of all archives: no sector > 98 for any target; S31/S88 off-silicon; S97 off SPOC footprint	CONFIRMED BLOCKED (no new epoch)
TIC 193718989	Localization robustness	TARGET_FAVORED_CONDITIONAL	15/14/7-sector difference images; injections on target and neighbours; broadened-PRF misspecification; flux-capable wide neighbours added	EXTENDED: catalogued neighbours excluded
TIC 193718989	Resolved ground / multiband	CONDITIONAL_LIMITS	193: ZTF DR24 re-analysed with propagated covariance, censoring and parity; 270/262: ZTF cannot observe these southern stars, ASAS-SN down	EXTENDED (193); BLOCKED (270, 262)
TIC 193718989	Planet / binary / brown dwarf / triple	EXCLUDED_SPECIFIED_CIRCULAR_LARGE_MODELS	Shape fits with timing nuisance; k-profiles; Gaia RV mass limits; MCMC radius	EXTENDED
TIC 193718989	Doubled period / eccentric	INCONCLUSIVE_GENERAL_ALTERNATIVES	Odd/even in 9 reductions; 2P twins via k-profile; e_min from ρ_{eff}; long-period box search	EXTENDED (262 disfavoured; 193 mild; 270 timing-dependent)
TIC 193718989	Secondary / injection recovery	NO_ROBUST_SECONDARY	Half-phase in 9 reductions with phase-shift null; 262 phase 0.387 re-measured	REPRODUCED: no robust secondary

Target	Test family	Prior status	FU3 action	FU3 outcome
TIC 193718989	Stellar / SED / Gaia	CONSISTENCY_ONLY	SED not re-fitted; Gaia DR3 flags added; colour-magnitude position against 71,717 stars within 100 pc (A4)	EXTENDED: 262 about 0.2 mag over-luminous (with RUWE 1.31, excess noise 15σ); 270, 193 not
TIC 193718989	Archival imaging	INCONCLUSIVE_CONTRAST	Not repeated: no new contrast data exist	NOT REPEATED
TIC 193718989	Spectra / RVs	BLOCKED_EPOCH_RVS	RAVE (262) and LAMOST (193) epochs located; Gaia RV constancy converted to limits	EXTENDED (summary-statistic limits)
TIC 193718989	Activity / orbital modulation	CONDITIONAL_CONSTRAINT	Not repeated	NOT REPEATED
TIC 193718989	FPP sensitivity	NOT_VALIDATED	Scenario families audited; image-conditioned bound; no new draws	AUDITED
TIC 270573718	Independent recurrence	RETROSPECTIVE_ONLY	Census of all archives: no sector > 98 for any target; S31/S88 off-silicon; S97 off SPOC footprint	CONFIRMED BLOCKED (no new epoch)
TIC 270573718	Localization robustness	TARGET_FAVORED_CONDITIONAL	15/14/7-sector difference images; injections on target and neighbours; broadened-PRF misspecification; flux-capable wide neighbours added	EXTENDED: catalogued neighbours excluded

Target	Test family	Prior status	FU3 action	FU3 outcome
TIC 270573718	Resolved ground / multiband	BLOCKED_DATA	193: ZTF DR24 re-analysed with propagated covariance, censoring and parity; 270/262: ZTF cannot observe these southern stars, ASAS-SN down	EXTENDED (193); BLOCKED (270, 262)
TIC 270573718	Planet / binary / brown dwarf / triple	EXCLUDED_SPECIFIED_CIRCULAR_LARGE_MODELS	Shape fits with timing nuisance; k-profiles; Gaia RV mass limits; MCMC radius	EXTENDED
TIC 270573718	Doubled period / eccentric	INCONCLUSIVE_GENERAL_ALTERNATIVES	Odd/even in 9 reductions; 2P twins via k-profile; e_min from ρ_{eff}; long-period box search	EXTENDED (262 disfavoured; 193 mild; 270 timing-dependent)
TIC 270573718	Secondary / injection recovery	NO_ROBUST_SECONDARY	Half-phase in 9 reductions with phase-shift null; 262 phase 0.387 re-measured	REPRODUCED: no robust secondary
TIC 270573718	Stellar / SED / Gaia	CONSISTENCY_ONLY	SED not re-fitted; Gaia DR3 flags added; colour-magnitude position against 71,717 stars within 100 pc (A4)	EXTENDED: 262 about 0.2 mag over-luminous (with RUWE 1.31, excess noise 15σ); 270, 193 not
TIC 270573718	Archival imaging	INCONCLUSIVE_CONTRAST	Not repeated: no new contrast data exist	NOT REPEATED
TIC 270573718	Spectra / RVs	BLOCKED_EPOCH_RVS	RAVE (262) and LAMOST (193) epochs located; Gaia RV constancy converted to limits	EXTENDED (summary-statistic limits)

Target	Test family	Prior status	FU3 action	FU3 outcome
TIC 270573718	Activity / orbital modulation	CONDITIONAL_CONSTRAINT	Not repeated	NOT REPEATED
TIC 270573718	FPP sensitivity	NOT_VALIDATED	Scenario families audited; image-conditioned bound; no new draws	AUDITED
TIC 262843927	Independent recurrence	RETROSPECTIVE_ONLY	Census of all archives: no sector > 98 for any target; S31/S88 off-silicon; S97 off SPOC footprint	CONFIRMED BLOCKED (no new epoch)
TIC 262843927	Localization robustness	TARGET_FAVORED_CONDITIONAL	15/14/7-sector difference images; injections on target and neighbours; broadened-PRF misspecification; flux-capable wide neighbours added	EXTENDED: catalogued neighbours excluded
TIC 262843927	Resolved ground / multiband	BLOCKED_DATA	193: ZTF DR24 re-analysed with propagated covariance, censoring and parity; 270/262: ZTF cannot observe these southern stars, ASAS-SN down	EXTENDED (193); BLOCKED (270, 262)
TIC 262843927	Planet / binary / brown dwarf / triple	EXCLUDED_SPECIFIED_CIRCULAR_LARGE_MODELS	Shape fits with timing nuisance; k-profiles; Gaia RV mass limits; MCMC radius	EXTENDED

Target	Test family	Prior status	FU3 action	FU3 outcome
TIC 262843927	Doubled period / eccentric	INCONCLUSIVE_GENERAL_ALTERNATIVES	Odd/even in 9 reductions; 2P twins via k-profile; e_min from ρ_{eff}; long-period box search	EXTENDED (262 disfavoured; 193 mild; 270 timing-dependent)
TIC 262843927	Secondary / injection recovery	NO_ROBUST_SECONDARY	Half-phase in 9 reductions with phase-shift null; 262 phase 0.387 re-measured	REPRODUCED: no robust secondary
TIC 262843927	Stellar / SED / Gaia	CONSISTENCY_ONLY	SED not re-fitted; Gaia DR3 flags added; colour-magnitude position against 71,717 stars within 100 pc (A4)	EXTENDED: 262 about 0.2 mag over-luminous (with RUWE 1.31, excess noise 15σ); 270, 193 not
TIC 262843927	Archival imaging	INCONCLUSIVE_CONTRAST	Not repeated: no new contrast data exist	NOT REPEATED
TIC 262843927	Spectra / RVs	BLOCKED_EPOCH_RVS	RAVE (262) and LAMOST (193) epochs located; Gaia RV constancy converted to limits	EXTENDED (summary-statistic limits)
TIC 262843927	Activity / orbital modulation	CONDITIONAL_CONSTRAINT	Not repeated	NOT REPEATED
TIC 262843927	FPP sensitivity	NOT_VALIDATED	Scenario families audited; image-conditioned bound; no new draws	AUDITED
all	Retrospective-split contradiction	prior: unexplained (270)	Exact reproduction and decomposition	RESOLVED (drift extrapolation)
all	Timestamp control (WASP-62 b)	new	FFI vs 2-min transit times, 9 sectors	FFI clock good to < 1 min

Target	Test family	Prior status	FU3 action	FU3 outcome
all	Empirical timing null	new	24 periodic injections per reduction	270 coherence p ≈ 0.04; 262/193 none
all	TGLC and FU3 pixel extractions	new	11 extra reductions	dip present in all
all	Flux budget, 1.5-3 px apertures	new	Gaia T mags, mission PRF	262 NFPP=0 aperture-dependent
all	Discovery replay	new	Inverted and injected search + Stage A	see Selection section

Failures, negative results and access limits

- The first replay launch ran its workers at below-normal priority, and they were starved next to the live survey (0-2 CPU-s in 35 min). It was killed and relaunched at normal priority. No result came from the first launch.
- A neighbour-selection defect (a G-magnitude cut) dropped TIC 193718989's 9.8″ star from the first localization run. It was fixed and rerun; the old outputs are kept as `*_run1_Gcut.json`.
- The first Gaia-RV limit used an invalid statistic for p_{obs} near 1. It was replaced by a relative statistic (amendment A2).
- Timing-null outputs initially overwrote one another. They were rerun per reduction; the stale files are renamed `*_STALE_*`.
- The PRF loader first picked up 33×33 “prf-phot” files on camera 4 CCD 2. It was fixed before any result was used.
- The covariance-weighted image statistic is unstable for some faint hosts (it went negative for one 262 neighbour). The diagonal-error statistic, which the injections calibrate, is the one reported.
- TGLC's S27 aperture event for TIC 270573718 reads 12 ppt, against 2-4 ppt in every other reduction and −1.4 ppt in TGLC PSF. It is retained in the tables and treated as a TGLC artefact.
- Unbounded single-source centroid fits wander up to about 100″ in low-S/N sectors. They are diagnostics only.
- The unbounded free-shape timing fit latched onto outliers in the unclipped pixel series (S1: a 25-ppt, 1.1-h fit at the duration bound). It is reported but not interpreted; a bounded, outlier-clipped variant was added post hoc (A3).
- The MCMC chains are short (autocorrelation 60-100 steps against 1,000 kept). Medians and 68% ranges are stable; 95% tails are approximate.

- Access: ExoFOP per-target endpoints return “Array”; the ASAS-SN light-curve service is down; the NASA Exoplanet Archive cone search was blocked (403); ZTF and ATLAS forced photometry are account-gated. None of these is a scientific negative.
- Negative results kept: every negative or zero event depth (event table); the half-phase nulls; the TIC 262843927 phase-0.387 null; the TIC 262843927 S31/S88 “no information” outcome; the non-significant timing χ^2 nulls.

Reproduction

Folder: `C:\Users\seanj\Documents\TESS_followup_262_193_270` (code, results, tables, figures, manifests, PROTOCOL.md); large caches in `D:\tess_fu3_cache` (TESScut cutouts, TGLC, PRF models, replay stores and runs). Interpreter: the tesshunter venv (python 3.13.3, numpy 2.5.1, scipy 1.18.0, astropy 8.0.1, batman 2.5.1, emcee 3.1.6, lightkurve 2.6.0).

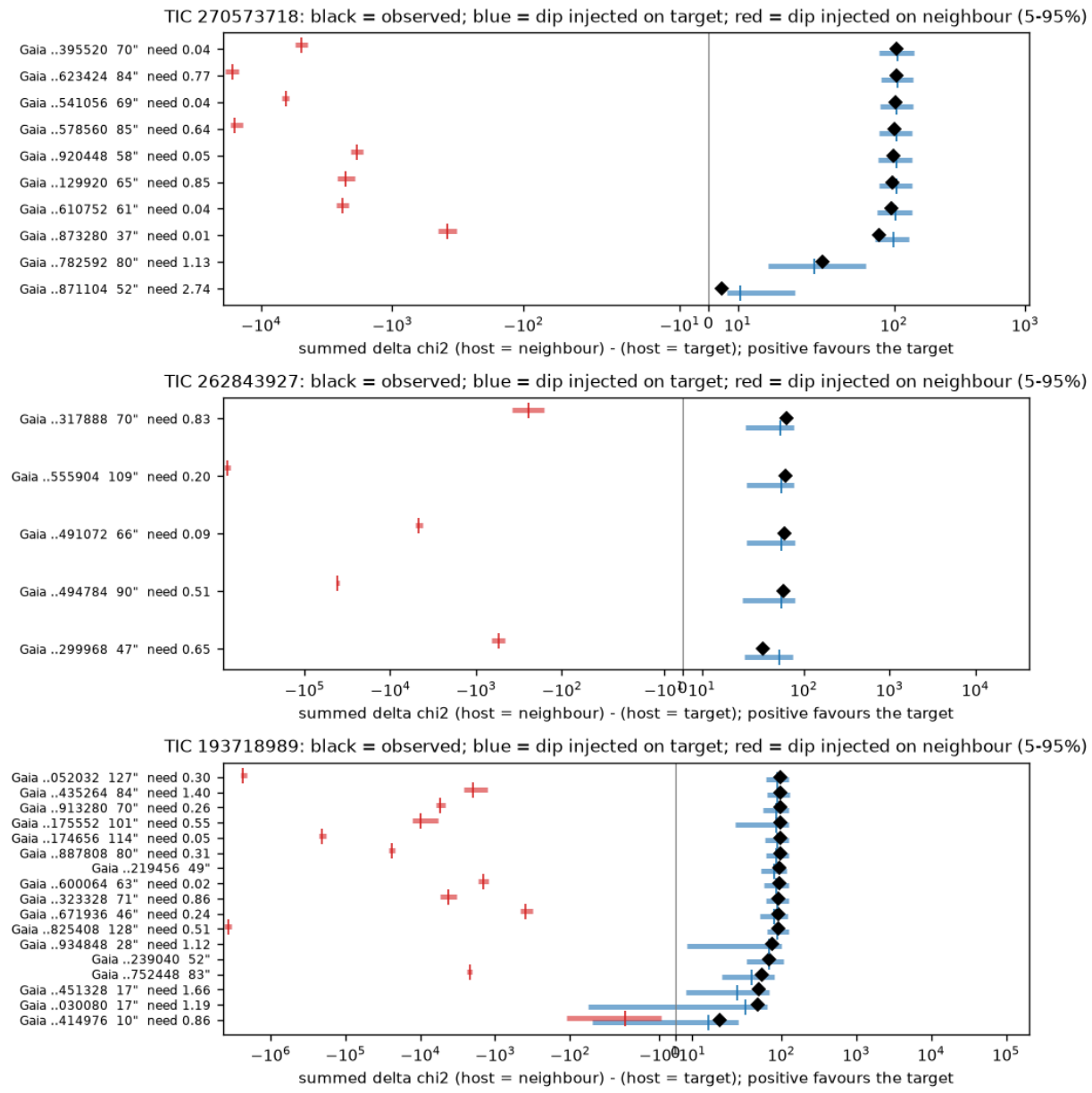
Order: `s00_inventory → s01_270_reproduce, s01b_decompose_all → s02_events → s03_calibrate → s04_oc → s05_tglc → s06_timestamp_control → s07a_cutouts → s07b_extract → s13_fluxbudget → s07c_localize → s09_262_s31_s88 → s10_secondary_2p, s10b_phase0387 → s11_shape → s12_fpp_audit → s14_ztf → s15_rv_limits → s16_ttv_null → s17_mcmc → s18_longp_bls → s19_freeshape_timing (plain and --bounded) → s20_gaia_epochs → s21_spoc2min_262 → s22_cmd_luminosity → s08_replay (select, build, run) → s08b_discovery_context → s08_summarize → facts, build_tables, make_figures → report_body (HTML) → make_pdf → make_manifest`. Every script writes JSON under `results/` and logs under `logs/`. Census scripts are under `code/census/`, with receipts in `results/census/receipts/`.

Machine-readable tables: `tables/event_table_all_reductions.csv` (every event in every reduction), pooled_depths, injection_null_calibration, timing_null_tests, model_comparison, host_localization, flux_budget, fpp_scenario_audit, secondary_odd_even, gaia_rv_limits, ztf_193718989, cmd_luminosity, freeshape_timing, discovery_funnel_targets and replay_summary (one row per replay search). Input hashes are in `manifests/input_manifest.csv`, the protocol hashes in `manifests/protocol_freeze.json`, the software versions in `manifests/environment.json`, and the size and SHA-256 of every output file in `manifests/output_manifest.csv`. A PDF copy of this report is `report/FU3_three_candidates.pdf`.

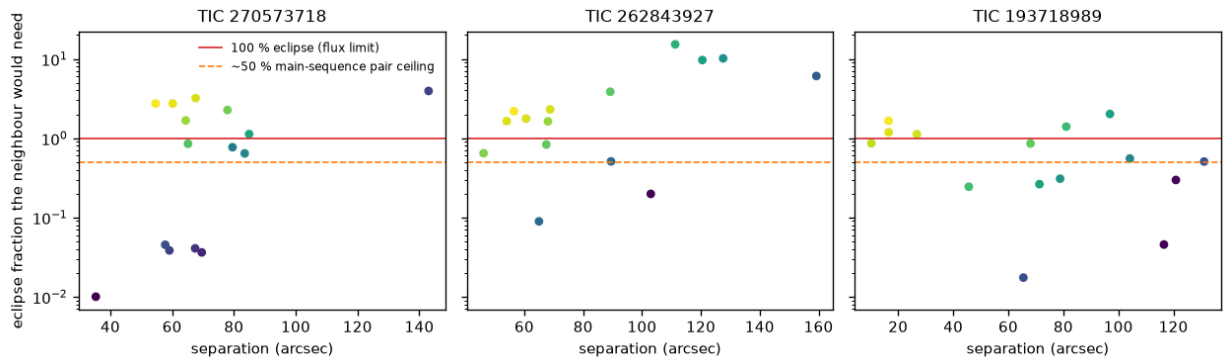
Unit tests: `code/tests/test_fu3.py`, 13 tests on synthetic data (event admission, unbiased depth and timing recovery, pooling, the Gaia-to-TESS magnitude relation, season grouping, the RV mass conversion, the replay's binomial limits

and the bounded free-shape fit). Run them with the venv interpreter: `python -m pytest code/tests -q`.

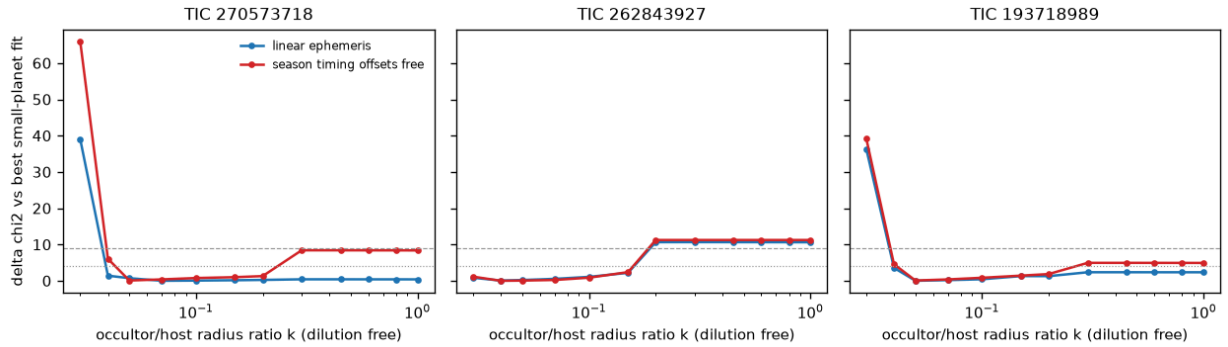
Summary figures



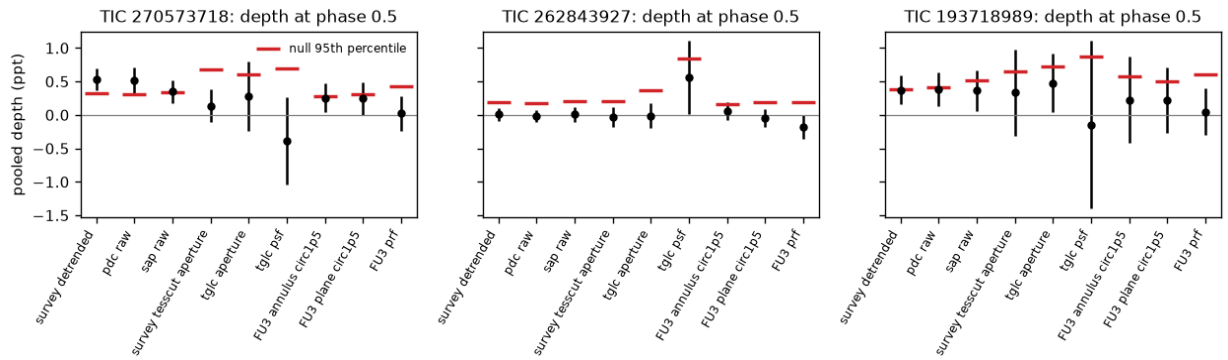
Injection-calibrated difference-image tests for every tested source. Black: observed summed $\Delta\chi^2$; blue: dips injected on the target (5-95%); red: dips injected on the neighbour (5-95%). Symmetric-log axis.



Eclipse fraction each catalogued neighbour would need (most favourable aperture 1.5-3 px and sector), against separation. Colour: ΔT.



Profile $\Delta\chi^2$ against occultor radius ratio with dilution free, for a linear ephemeris and with per-season timing offsets.



Depth at phase 0.5 in each reduction (black) with the 95th percentile of 300 phase-shifted null measurements (red).

Built 2026-10-19T06:00:21Z from FU3 results. Conditional analysis; no validation claim.